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TOP 10 TAKE-HOME MESSAGES FOR THE EVALUATION AND DIAGNOSIS OF CHEST PAIN

- 1. Chest Pain Means More Than Pain in the Chest.** Pain, pressure, tightness, or discomfort in the chest, shoulders, arms, neck, back, upper abdomen, or jaw, as well as shortness of breath and fatigue should all be considered anginal equivalents.
- 2. High-Sensitivity Troponins Preferred.** High-sensitivity cardiac troponins are the preferred standard for establishing a biomarker diagnosis of acute myocardial infarction, allowing for more accurate detection and exclusion of myocardial injury.
- 3. Early Care for Acute Symptoms.** Patients with acute chest pain or chest pain equivalent symptoms should seek medical care immediately by calling 9-1-1. Although most patients will not have a cardiac cause, the evaluation of all patients should focus on the early identification or exclusion of life-threatening causes.
- 4. Share the Decision-Making.** Clinically stable patients presenting with chest pain should be included in decision-making; information about risk

of adverse events, radiation exposure, costs, and alternative options should be provided to facilitate the discussion.

- 5. Testing Not Needed Routinely for Low-Risk Patients.** For patients with acute or stable chest pain determined to be low risk, urgent diagnostic testing for suspected coronary artery disease is not needed.
- 6. Pathways.** Clinical decision pathways for chest pain in the emergency department and outpatient settings should be used routinely.
- 7. Accompanying Symptoms.** Chest pain is the dominant and most frequent symptom for both men and women ultimately diagnosed with acute coronary syndrome. Women may be more likely to present with accompanying symptoms such as nausea and shortness of breath.
- 8. Identify Patients Most Likely to Benefit From Further Testing.** Patients with acute or stable chest pain who are at intermediate risk or intermediate to high pre-test risk of obstructive coronary artery disease, respectively, will benefit the most from cardiac imaging and testing.
- 9. Noncardiac Is In. Atypical Is Out.** "Noncardiac" should be used if heart disease is not suspected. "Atypical" is a misleading descriptor of chest pain, and its use is discouraged.
- 10. Structured Risk Assessment Should Be Used.** For patients presenting with acute or stable chest pain, risk for coronary artery disease and adverse events should be estimated using evidence-based diagnostic protocols.

Figure 1 illustrates the take-home messages.

PREAMBLE

Since 1980, the American College of Cardiology (ACC) and American Heart Association (AHA) have translated scientific evidence into clinical practice guidelines with recommendations to improve cardiovascular health. These guidelines, which are based on systematic methods to evaluate and classify evidence, provide a foundation for the delivery of quality cardiovascular care. The ACC and AHA sponsor the development and publication of clinical practice guidelines without commercial support, and members volunteer their time to the writing and review efforts. Guidelines are official policy of the ACC and AHA. For some guidelines, the ACC and AHA partner with other organizations.

Intended Use

Clinical practice guidelines provide recommendations applicable to patients with or at risk of developing cardiovascular disease. The focus is on medical practice

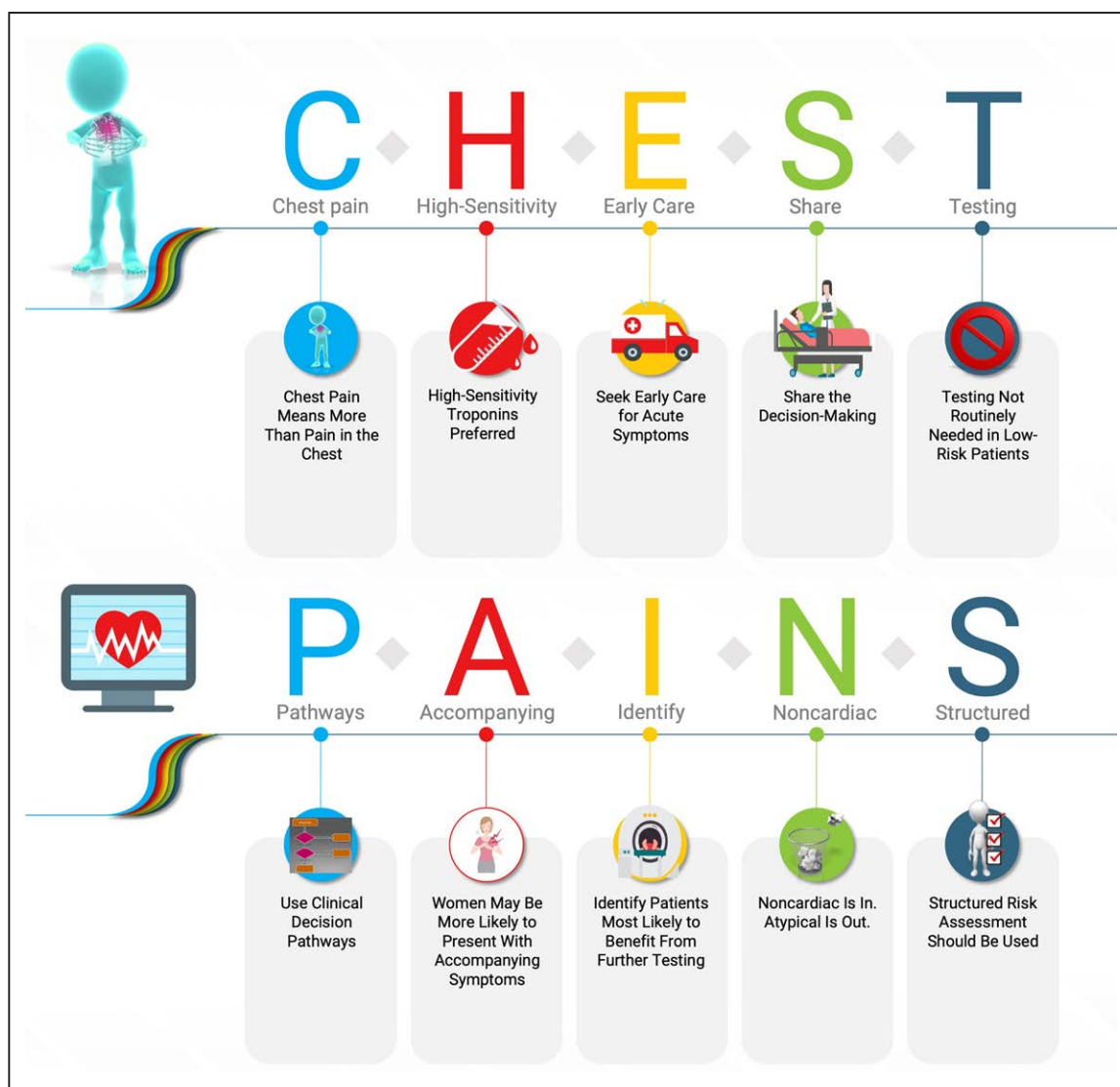


Figure 1. Take-Home Messages for the Evaluation and Diagnosis of Chest Pain

in the United States, but these guidelines are relevant to patients throughout the world. Although guidelines may be used to inform regulatory or payer decisions, the intent is to improve quality of care and align with patients' interests. Guidelines are intended to define practices meeting the needs of patients in most, but not all, circumstances and should not replace clinical judgment.

Clinical Implementation

Management, in accordance with guideline recommendations, is effective only when followed by both practitioners and patients. Adherence to recommendations can be enhanced by shared decision-making between clinicians and patients, with patient engagement in selecting interventions on the basis of individual values, preferences, and associated conditions and comorbidities.

Methodology and Modernization

The ACC/AHA Joint Committee on Clinical Practice Guidelines (Joint Committee) continuously reviews, updates, and modifies guideline methodology on the basis of published standards from organizations, including the Institute of Medicine,^{1,2} and on the basis of internal reevaluation. Similarly, presentation and delivery of guidelines are reevaluated and modified in response to evolving technologies and other factors to optimally facilitate dissemination of information to healthcare professionals at the point of care.

Numerous modifications to the guidelines have been implemented to make them shorter and enhance "user friendliness." Guidelines are written and presented in a modular, "knowledge chunk" format, in which each chunk includes a table of recommendations, a brief synopsis, recommendation-specific supportive text and, when

Table 2. Associated Guidelines and Statements

Title	Organization	Publication Year (Reference)
Guidelines		
Stable ischemic heart disease	ACC/AHA/AATS/PCNA/SCAI/STS	2014 ^{1*} 2012 ²
Atrial fibrillation	AHA/ACC/HRS	2014 ^{3*} 2019 ⁴
Non-ST elevation ACS	AHA/ACC	2014 ⁵
Blood cholesterol	AHA/ACC/AACVPR/AAPA/ABC/ACPM/ADA/AGS/APhA/ASPC/NLA/PCNA	2018 ⁶
Heart failure	ACC/AHA	2013 ^{7*} 2017 ⁸
Primary prevention of cardiovascular disease	ACC/AHA	2019 ⁹
Management of overweight and obesity in adults	AHA/ACC/TOS	2014 ¹⁰
ST-elevation myocardial infarction	ACC/AHA	2013 ¹¹
Ventricular arrhythmias and the prevention of sudden cardiac death	AHA/ACC/HRS	2017 ¹²
Coronary artery bypass graft surgery	ACC/AHA	2011 ¹³
Hypertrophic cardiomyopathy	ACC/AHA	2020 ¹⁴
Percutaneous coronary intervention	ACC/AHA/SCAI	2011 ^{15*} 2015 ¹⁶
Secondary prevention and risk reduction therapy for patients with coronary and other atherosclerotic vascular disease	AHA/ACC	2011 ¹⁷
Guidelines for cardiopulmonary resuscitation and emergency cardiovascular care—part 9: postcardiac arrest care	AHA	2010 ^{18*} 2019 ¹⁹
Prevention, detection, evaluation, and management of high blood pressure in adults	ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA	2017 ²⁰
Statements		
Testing of low-risk patients presenting to the emergency department with chest pain	AHA	2010 ²¹
Prevention of cardiovascular disease in adults with type 2 diabetes mellitus	AHA/ADA	2015 ²²
Prevention and control of seasonal influenza with vaccines	CDC	2018 ²³

AACVPR indicates American Association of Cardiovascular and Pulmonary Rehabilitation; AAPA, American Academy of Physician Assistants; AATS, American Association for Thoracic Surgery; ABC, Association of Black Cardiologists; ACC, American College of Cardiology; ACPM, American College of Preventive Medicine; ADA, American Diabetes Association; AGS, American Geriatrics Society; AHA, American Heart Association; APhA, American Pharmacists Association; ASH, American Society of Hypertension; ASPC, American Society for Preventive Cardiology; CDC, Centers for Disease Control and Prevention; ESC, European Society of Cardiology; HRS, Heart Rhythm Society; NHLBI, National Heart, Lung, and Blood Institute; NLA, National Lipid Association; NMA, National Medical Association; PCNA, Preventive Cardiovascular Nurses Association; SCAI, Society for Cardiovascular Angiography and Interventions; STS, Society of Thoracic Surgeons; and TOS, The Obesity Society.

*The full-text guideline and focused update references are provided.

1.3. Document Review and Approval

This document was reviewed by 16 official reviewers nominated by the ACC, the American College of Emergency Physicians, AHA, ASE, American Society of Nuclear Cardiology, CHEST, SAEM, SCCT, and SCMR, and 39 individual content reviewers. Reviewers' RWI information was distributed to the writing committee and is published in this document (Appendix 2).

This document was approved for publication by the governing bodies of the ACC and the AHA and was endorsed by the ASE, CHEST, SAEM, SCCT, and SCMR.

1.4. Scope of the Guideline

The charge of the writing committee was to develop a guideline for the evaluation of acute or stable chest pain or other anginal equivalents, in various clinical settings, with an

emphasis on the diagnosis on ischemic causes. This guideline will not provide recommendations on whether revascularization is appropriate or what modality is indicated. Such recommendations can be found in the forthcoming ACC/AHA coronary artery revascularization guideline. In developing the "2021 AHA/ACC/ASE/CHEST/SAEM/SCCT/SCMR Guideline for the Evaluation and Diagnosis of Chest Pain," the writing committee first reviewed previous published guidelines and related statements. Table 2 contains a list of these publications deemed pertinent to this writing effort and is intended for use as a resource, thus obviating the need to repeat existing guideline recommendations.

1.4.1. Scope of the Problem

Synopsis

After injuries, chest pain is the second most common reason for adults to present to the emergency

department (ED) in the United States and accounts for >6.5 million visits, which is 4.7% of all ED visits.¹ Chest pain also leads to nearly 4 million outpatient visits annually in the United States.² Chest pain remains a diagnostic challenge in the ED and outpatient setting and requires thorough clinical evaluation. Although the cause of chest pain is often noncardiac, coronary artery disease (CAD) affects >18.2 million adults in the United States and remains the leading cause of death for men and women, accounting for >365 000 deaths annually.³ Distinguishing between serious and benign causes of chest pain is imperative. The lifetime prevalence of chest pain in the United States is 20% to 40%,⁴ and women experience this symptom more often than men.⁵ Of all ED patients with chest pain, only 5.1% will have an acute coronary syndrome (ACS), and more than half will ultimately be found to have a noncardiac cause.⁶ Nonetheless, chest pain is the most common symptom of CAD in both men and women.

1.4.2. Defining Chest Pain

Recommendations for Defining Chest Pain Referenced studies that support the recommendations are summarized in Online Data Supplements 1 and 2.		
COR	LOE	Recommendations
1	B-NR	1. An initial assessment of chest pain is recommended to triage patients effectively on the basis of the likelihood that symptoms may be attributable to myocardial ischemia. ¹⁻⁷
1	C-LD	2. Chest pain should not be described as atypical, because it is not helpful in determining the cause and can be misinterpreted as benign in nature. Instead, chest pain should be described as cardiac, possibly cardiac, or noncardiac because these terms are more specific to the potential underlying diagnosis.

Synopsis

Chest pain is one of the most common reasons that people seek medical care. The term “chest pain” is used by patients and applied by clinicians to describe the many unpleasant or uncomfortable sensations in the anterior chest that prompt concern for a cardiac problem. Chest pain should be considered *acute* when it is new onset or involves a change in pattern, intensity, or duration compared with previous episodes in a patient with recurrent symptoms. Chest pain should be considered *stable* when symptoms are chronic and associated with consistent precipitants such as exertion or emotional stress.

Although the term chest pain is used in clinical practice, patients often report pressure, tightness, squeezing, heaviness, or burning. In this regard, a more appropriate term is “chest discomfort,” because patients may not use the descriptor “pain.” They may also report a location other than the chest, including the shoulder, arm, neck, back, upper abdomen, or jaw. Despite individual variability, the discomfort induced by myocardial ischemia is often characteristic and therefore central to the diagnosis. For this reason, features more likely to be associated with ischemia have been described as typical versus atypical; however, the latter can be confusing because it is frequently used to describe symptoms considered nonischemic as well as noncardiac. Although other nonclassic symptoms of ischemia, such as shortness of breath, nausea, radiating discomfort, or numbness, may be present, chest pain or chest discomfort remains the predominant symptom reported in men and women who are ultimately diagnosed with myocardial ischemia.³⁻⁷ Pain—described as sharp, fleeting, related to inspiration (pleuritic) or position, or shifting locations—suggests a lower likelihood of ischemia.

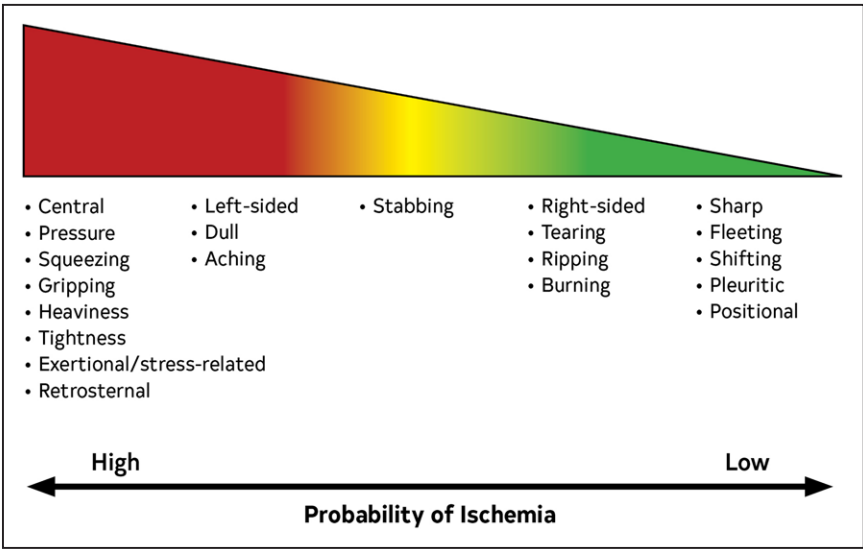


Figure 2. Index of Suspicion That Chest “Pain” Is Ischemic in Origin on the Basis of Commonly Used Descriptors

Recommendation-Specific Supportive Text

1. Like most visceral discomfort, the sensation produced by myocardial ischemia is characteristically deep, difficult to localize, and usually diffuse. Point tenderness renders ischemia less likely. Reported symptoms lie somewhere on a continuum of higher or lower probability of ischemia based on the presence or absence of specific characteristics (Figure 2). Other clinical elements (eg, duration, provoking and relieving factors, patient age, cardiac risk factors) provide further focus toward or away from ischemia in the diagnostic process. It is essential to ascertain the characteristics of the chest pain directly from the patient for optimal interpretation.¹⁻⁷ A patient's history is the most important basis for considering presence or absence of myocardial ischemia, but the source of cardiac symptoms is complex, and their expression is variable. The diagnosis of ischemia may require data beyond history alone. In some patients, what appears to be noncardiac chest pain may be ischemic in origin.
2. Chest pain has been traditionally stratified into "typical" and "atypical" types. Chest pain that is more likely associated with ischemia consists of substernal chest discomfort provoked by exertion or emotional stress and relieved by rest or nitroglycerin. The more classic the chest discomfort is based on quality, location, radiation, and provoking and relieving factors, the more likely it is to be of cardiac ischemic origin. Atypical chest pain is a problematic term. Although it was intended to indicate angina without typical chest symptoms, it is more often used to state that the symptom is noncardiac in origin. As such, we discourage the use of atypical chest pain. Emphasis is more constructively placed on specific aspects of symptoms that suggest their origin in terms of probable ischemia. Of note, chest pain is broadly defined to also include referred pain in the shoulders, arms, jaw, neck, and upper abdomen. To diminish ambiguity, use "cardiac," "possible cardiac," and "noncardiac" to describe the suspected cause of chest pain is encouraged.

1.5. Abbreviations

Abbreviation	Meaning/Phrase
ACS	acute coronary syndrome
AMI	acute myocardial infarction
CABG	coronary artery bypass graft
CAC	coronary artery calcium
CAD	coronary artery disease
CCTA	coronary computed tomographic angiography
CDP	clinical decision pathway
CMR	cardiovascular magnetic resonance
cTn	cardiac troponin
ECG	electrocardiogram
ED	emergency department
EMS	emergency medical services
FFR-CT	fractional flow reserve with computed tomography
GDMT	guideline-directed medical therapy
hs-cTn	high-sensitivity cardiac troponin
ICA	invasive coronary angiography
INOCA	ischemia and no obstructive coronary artery disease
MACE	major adverse cardiovascular events
MBFR	myocardial blood flow reserve
METs	metabolic equivalents
MINOCA	myocardial infarction and nonobstructive coronary arteries
MPI	myocardial perfusion imaging
NSTE-ACS	non-ST-segment-elevation acute coronary syndrome
PCI	percutaneous coronary intervention
PE	pulmonary embolism
PET	positron emission tomography
SIHD	stable ischemic heart disease
SPECT	single-photon emission computed tomography
STEMI	ST-segment-elevation myocardial infarction
TEE	transesophageal echocardiography
TTE	transthoracic echocardiography
VF	ventricular fibrillation
VHD	valvular heart disease
VT	ventricular tachycardia

15. Levine GN, Bates ER, Blankenship JC, et al. 2011 ACCF/AHA/SCAI guideline for percutaneous coronary intervention: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines and the Society for Cardiovascular Angiography and Interventions. *Circulation*. 2011;124:e574–e651.
16. Levine GN, Bates ER, Blankenship JC, et al. 2015 ACC/AHA/SCAI focused update on primary percutaneous coronary intervention for patients with ST-elevation myocardial infarction: an update of the 2011 ACCF/AHA/SCAI guideline for percutaneous coronary intervention and the 2013 ACCF/AHA guideline for the management of ST-elevation myocardial infarction. *Circulation*. 2016;133:1135–1147.
17. Smith SC Jr, Benjamin EJ, Bonow RO, et al. AHA/ACCF secondary prevention and risk reduction therapy for patients with coronary and other atherosclerotic vascular disease: 2011 update: a guideline from the American Heart Association and American College of Cardiology Foundation. *Circulation*. 2011;124:2458–2473.
18. Berg KM, Cheng A, Panchal AR, et al. Part 7: systems of care: 2020 American Heart Association guidelines for cardiopulmonary resuscitation and emergency cardiovascular care. *Circulation*. 2020;142:S580–S604.
19. Panchal AR, Berg KM, Hirsch KG, et al. 2019 American Heart Association focused update on advanced cardiovascular life support: use of advanced airways, vasopressors, and extracorporeal cardiopulmonary resuscitation during cardiac arrest: an update to the American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. *Circulation*. 2019;140:e881–e894.
20. Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Hypertension*. 2018;71:e13–e115.
21. Amsterdam EA, Kirk JD, Bluemke DA, et al. Testing of low-risk patients presenting to the emergency department with chest pain: a scientific statement from the American Heart Association. *Circulation*. 2010;122:1756–1776.
22. Fox CS, Golden SH, Anderson C, et al. Update on prevention of cardiovascular disease in adults with type 2 diabetes mellitus in light of recent evidence: a scientific statement from the American Heart Association and the American Diabetes Association. *Circulation*. 2015;132:691–718.
23. Grohskopf LA, Sokolow LZ, Broder KR, et al. Prevention and control of seasonal influenza with vaccines: recommendations of the Advisory Committee on Immunization Practices—United States, 2018–19 Influenza Season. *MMWR Recomm Rep*. 2018;67:1–20.

1.4.1. Scope of the Problem

1. Rui P, Kang K. National Hospital Ambulatory Medical Care Survey: 2017 emergency department summary tables. National Center for Health Statistics. Available at: https://www.cdc.gov/nchs/data/nhamcs/web_tables/2017_ed_web_tables-508.pdf. Accessed February 12, 2021.
2. Rui P, Okeyode T. National Ambulatory Medical Care Survey: 2016 national summary tables. 2016 Available at: https://www.cdc.gov/nchs/data/ahcd/namcs_summary/2016_namcs_web_tables.pdf. Accessed February 12, 2021.
3. Virani SS, Alonso A, Benjamin EJ, et al. Heart disease and stroke statistics-2020 update: a report from the American Heart Association. *Circulation*. 2020;141:e139–e596.
4. Ruigomez A, Rodriguez LA, Wallander MA, et al. Chest pain in general practice: incidence, comorbidity and mortality. *Fam Pract*. 2006;23:167–174.
5. Bosner S, Becker A, Haasensitter J, et al. Chest pain in primary care: epidemiology and pre-work-up probabilities. *Eur J Gen Pract*. 2009;15:141–146.
6. Hsia RY, Hale Z, Tabas JA. A national study of the prevalence of life-threatening diagnoses in patients with chest pain. *JAMA Intern Med*. 2016;176:1029–1032.

1.4.2. Defining Chest Pain

1. Canto JG, Rogers WJ, Goldberg RJ, et al. Association of age and sex with myocardial infarction symptom presentation and in-hospital mortality. *JAMA*. 2012;307:813–822.
2. van der Meer MG, Backus BE, van der Graaf Y, et al. The diagnostic value of clinical symptoms in women and men presenting with chest pain at the emergency department, a prospective cohort study. *PLoS One*. 2015;10:e0116431.
3. Hemal K, Pagidipati NJ, Coles A, et al. Sex differences in demographics, risk factors, presentation, and noninvasive testing in stable outpatients with

suspected coronary artery disease: insights from the PROMISE trial. *J Am Coll Cardiol Img*. 2016;9:337–346.

4. Lichtman JH, Leifheit EC, Safdar B, et al. Sex differences in the presentation and perception of symptoms among young patients with myocardial infarction: evidence from the VIRGO Study (Variation in Recovery: Role of Gender on Outcomes of Young AMI Patients). *Circulation*. 2018;137:781–790.
5. Bosner S, Haasensitter J, Hani MA, et al. Gender differences in presentation and diagnosis of chest pain in primary care. *BMC Fam Pract*. 2009;10:79.
6. Ferry AV, Anand A, Strachan FE, et al. Presenting symptoms in men and women diagnosed with myocardial infarction using sex-specific criteria. *J Am Heart Assoc*. 2019;8:e012307.
7. Kreatsoula C, Fleegler EW, Kubzansky LD, et al. Young adults and adverse childhood events: a potent measure of cardiovascular risk. *Am J Med*. 2019;132:605–613.

2. INITIAL EVALUATION

2.1. History

1. Amsterdam EA, Wenger NK, Brindis RG, et al. 2014 AHA/ACC guideline for the management of patients with non-ST-elevation acute coronary syndromes: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2014;130:e344–e426.
2. Lindsell CJ, Anantharaman V, Diercks D, et al. The Internet Tracking Registry of Acute Coronary Syndromes (iTrACS): a multicenter registry of patients with suspicion of acute coronary syndromes reported using the standardized reporting guidelines for emergency department chest pain studies. *Ann Emerg Med*. 2006;48:666–677; 77 e1–9.
3. Hsia RY, Hale Z, Tabas JA. A national study of the prevalence of life-threatening diagnoses in patients with chest pain. *JAMA Intern Med*. 2016;176:1029–1032.
4. Fanaroff AC, Rymer JA, Goldstein SA, et al. Does this patient with chest pain have acute coronary syndrome?: the rational clinical examination systematic review. *JAMA*. 2015;314:1955–1965.
5. Diercks DB, Boghos E, Guzman H, et al. Changes in the numeric descriptive scale for pain after sublingual nitroglycerin do not predict cardiac etiology of chest pain. *Ann Emerg Med*. 2005;45:581–585.

2.1.1. A Focus on the Uniqueness of Chest Pain in Women

1. Hemal K, Pagidipati NJ, Coles A, et al. Sex differences in demographics, risk factors, presentation, and noninvasive testing in stable outpatients with suspected coronary artery disease: insights from the PROMISE trial. *J Am Coll Cardiol Img*. 2016;9:337–346.
2. Garcia M, Mulvagh SL, Merz CN, et al. Cardiovascular disease in women: clinical perspectives. *Circ Res*. 2016;118:1273–1293.
3. Lichtman JH, Leifheit EC, Safdar B, et al. Sex differences in the presentation and perception of symptoms among young patients with myocardial infarction: evidence from the VIRGO Study (Variation in Recovery: Role of Gender on Outcomes of Young AMI Patients). *Circulation*. 2018;137:781–790.
4. Pelletier R, Khan NA, Cox J, et al. Sex versus gender-related characteristics: which predicts outcome after acute coronary syndrome in the young? *J Am Coll Cardiol*. 2016;67:127–135.
5. Ferry AV, Anand A, Strachan FE, et al. Presenting symptoms in men and women diagnosed with myocardial infarction using sex-specific criteria. *J Am Heart Assoc*. 2019;8:e012307.
6. Kreatsoula C, Dinakar D, Mehta S, et al. Machine learning to evaluate gender differences in typical and atypical angina among patients with obstructive coronary artery disease. ESC Congress 2019. Paris, France. 2019.
7. DeFilippis EM, Collins BL, Singh A, et al. Women who experience a myocardial infarction at a young age have worse outcomes compared with men: the Mass General Brigham YOUNG-MI registry. *Eur Heart J*. 2020;41:4127–4137.
8. Rui P, Kang K, Ashman JJ. National Hospital Ambulatory Medical Care Survey: 2016 Emergency Department Summary Tables. 2016 Available at: https://www.cdc.gov/nchs/data/ahcd/nhamcs_emergency/2016_ed_web_tables.pdf. Accessed July 26, 2021.
9. Reynolds HR, Shaw LJ, Min JK, et al. Association of sex with severity of coronary artery disease, ischemia, and symptom burden in patients with

moderate or severe ischemia: secondary analysis of the ISCHEMIA randomized clinical trial. *JAMA Cardiol.* 2020;5:1–14.

10. Meisel ZF, Armstrong K, Mechem CC, et al. Influence of sex on the out-of-hospital management of chest pain. *Acad Emerg Med.* 2010;17:80–87.
11. Khan NA, Daskalopoulou SS, Karp I, et al. Sex differences in prodromal symptoms in acute coronary syndrome in patients aged 55 years or younger. *Heart.* 2017;103:863–869.
12. Roger VL, Farkouh ME, Weston SA, et al. Sex differences in evaluation and outcome of unstable angina. *JAMA.* 2000;283:646–652.
13. McSweeney JC, Cleves MA, Zhao W, et al. Cluster analysis of women's prodromal and acute myocardial infarction symptoms by race and other characteristics. *J Cardiovasc Nurs.* 2010;25:311–322.
14. Safdar B, Nagurney JT, Anise A, et al. Gender-specific research for emergency diagnosis and management of ischemic heart disease: proceedings from the 2014 Academic Emergency Medicine Consensus Conference Cardiovascular Research Workgroup. *Acad Emerg Med.* 2014;21:1350–1360.
15. Khan NA, Daskalopoulou SS, Karp I, et al. Sex differences in acute coronary syndrome symptom presentation in young patients. *JAMA Intern Med.* 2013;173:1863–1871.
16. Tamis-Holland JE, Lu J, Korytkowski M, et al. Sex differences in presentation and outcome among patients with type 2 diabetes and coronary artery disease treated with contemporary medical therapy with or without prompt revascularization: a report from the BARI 2D Trial (Bypass Angioplasty Revascularization Investigation 2 Diabetes). *J Am Coll Cardiol.* 2013;61:1767–1776.

2.1.2. Considerations for Older Patients With Chest Pain

1. Grosmaître P, Le Vasseur O, Yachouh E, et al. Significance of atypical symptoms for the diagnosis and management of myocardial infarction in elderly patients admitted to emergency departments. *Arch Cardiovasc Dis.* 2013;106:586–592.
2. Jokhadar M, Wenger NK. Review of the treatment of acute coronary syndrome in elderly patients. *Clin Interv Aging.* 2009;4:435–444.
3. Gupta R, Munoz R. Evaluation and management of chest pain in the elderly. *Emerg Med Clin North Am.* 2016;34:523–542.
4. Lowenstern A, Alexander KP, Hill CL, et al. Age-related differences in the noninvasive evaluation for possible coronary artery disease: insights from the Prospective Multicenter Imaging Study for Evaluation of Chest Pain (PROMISE) Trial. *JAMA Cardiol.* 2020;5:193–201.

2.1.3. Considerations for Diverse Patient Populations With Chest Pain

1. Virani SS, Alonso A, Benjamin EJ, et al. Heart disease and stroke statistics-2020 update: a report from the American Heart Association. *Circulation.* 2020;141:e139–e596.
2. DeVon HA, Burke LA, Nelson H, et al. Disparities in patients presenting to the emergency department with potential acute coronary syndrome: it matters if you are Black or White. *Heart Lung.* 2014;43:270–277.
3. Alrwisan A, Eworuke E. Are discrepancies in waiting time for chest pain at emergency departments between African Americans and Whites improving over time? *J Emerg Med.* 2016;50:349–355.
4. Graham G. Racial and ethnic differences in acute coronary syndrome and myocardial infarction within the United States: from demographics to outcomes. *Clin Cardiol.* 2016;39:299–306.
5. Lopez L, Wilper AP, Cervantes MC, et al. Racial and sex differences in emergency department triage assessment and test ordering for chest pain, 1997–2006. *Acad Emerg Med.* 2010;17:801–808.
6. Mozaffarian D, Benjamin EJ, Go AS, et al. Heart disease and stroke statistics-2015 update: a report from the American Heart Association. *Circulation.* 2015;131:e29–e322.
7. Olson MB, Kelsey SF, Matthews K, et al. Symptoms, myocardial ischaemia and quality of life in women: results from the NHLBI-sponsored WISE Study. *Eur Heart J.* 2003;24:1506–1514.
8. Volgman AS, Palaniappan LS, Aggarwal NT, et al. Atherosclerotic cardiovascular disease in South Asians in the United States: epidemiology, risk factors, and treatments: a scientific statement from the American Heart Association. *Circulation.* 2018;138:e1–e34.
9. Kendall H, Marley A, Patel JV, et al. Hospital delay in South Asian patients with acute ST-elevation myocardial infarction in the UK. *Eur J Prev Cardiol.* 2013;20:737–742.

10. Zaman MJ, Junghans C, Sekhri N, et al. Presentation of stable angina pectoris among women and South Asian people. *CMAJ.* 2008;179:659–667.
11. King-Shier K, Quan H, Kapral MK, et al. Acute coronary syndromes presentations and care outcomes in white, South Asian and Chinese patients: a cohort study. *BMJ Open.* 2019;9:e022479.

2.1.4. Patient-Centric Considerations

1. Leifheit-Limson EC, D'Onofrio G, Daneshvar M, et al. Sex differences in cardiac risk factors, perceived risk, and health care provider discussion of risk and risk modification among young patients with acute myocardial infarction: the VIRGO Study. *J Am Coll Cardiol.* 2015;66:1949–1957.
2. Mathews R, Peterson ED, Li S, et al. Use of emergency medical service transport among patients with ST-segment-elevation myocardial infarction: findings from the National Cardiovascular Data Registry Acute Coronary Treatment Intervention Outcomes Network Registry-Get With The Guidelines. *Circulation.* 2011;124:154–163.
3. Becker L, Larsen MP, Eisenberg MS. Incidence of cardiac arrest during self-transport for chest pain. *Ann Emerg Med.* 1996;28:612–616.
4. Deputy Heart Attack Program with Early Heart Attack Care Education. Available at: <https://dha.aacc.org/>. Accessed September 10, 2020.
5. Merchant RM, Topjian AA, Panchal AR, et al. Part 1: executive summary: 2020 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. *Circulation.* 2020;142:S337–S357.

2.2. Physical Examination

1. Fanaroff AC, Rymer JA, Goldstein SA, et al. Does this patient with chest pain have acute coronary syndrome?: the rational clinical examination systematic review. *JAMA.* 2015;314:1955–1965.
2. McConaghy JR, Oza RS. Outpatient diagnosis of acute chest pain in adults. *Am Fam Physician.* 2013;87:177–182.
3. Ohle R, Um J, Anjum O, et al. High risk clinical features for acute aortic dissection: a case-control study. *Acad Emerg Med.* 2018;25:378–387.
4. Garas G, Zarogoulidis P, Efthymiou A, et al. Spontaneous esophageal rupture as the underlying cause of pneumothorax: early recognition is crucial. *J Thorac Dis.* 2014;6:1655–1658.
5. Grani C, Senn O, Bischof M, et al. Diagnostic performance of reproducible chest wall tenderness to rule out acute coronary syndrome in acute chest pain: a prospective diagnostic study. *BMJ Open.* 2015;5:e007442.
6. Panju AA, Hemmelgarn BR, Guyatt GH, et al. The rational clinical examination. Is this patient having a myocardial infarction? *JAMA.* 1998;280:1256–1263.
7. Perrier A, Desmarais S, Miron MJ, et al. Non-invasive diagnosis of venous thromboembolism in outpatients. *Lancet.* 1999;353:190–195.
8. Tsai TT, Trimarchi S, Nienaber CA. Acute aortic dissection: perspectives from the International Registry of Acute Aortic Dissection (IRAD). *Eur J Vasc Endovasc Surg.* 2009;37:149–159.
9. von Kodolitsch Y, Schwartz AG, Nienaber CA. Clinical prediction of acute aortic dissection. *Arch Intern Med.* 2000;160:2977–2982.
10. Klompas M. Does this patient have an acute thoracic aortic dissection? *JAMA.* 2002;287:2262–2272.

2.3. Diagnostic Testing

2.3.1. Setting Considerations

1. Diercks DB, Kontos MC, Chen AY, et al. Utilization and impact of pre-hospital electrocardiograms for patients with acute ST-segment elevation myocardial infarction: data from the NCDR (National Cardiovascular Data Registry) ACTION (Acute Coronary Treatment and Intervention Outcomes Network) Registry. *J Am Coll Cardiol.* 2009;53:161–166.
2. Puymirat E, Simon T, Steg PG, et al. Association of changes in clinical characteristics and management with improvement in survival among patients with ST-elevation myocardial infarction. *JAMA.* 2012;308:998–1006.
3. Jollis JG, Granger CB, Henry TD, et al. Systems of care for ST-segment-elevation myocardial infarction: a report from the American Heart Association's Mission: Lifeline. *Circ Cardiovasc Qual Outcomes.* 2012;5:423–428.
4. Postma S, Bergmeijer T, ten Berg J, et al. Pre-hospital diagnosis, triage and treatment in patients with ST elevation myocardial infarction. *Heart.* 2012;98:1674–1678.

5. Patel M, Dunford JV, Aguilar S, et al. Pre-hospital electrocardiography by emergency medical personnel: effects on scene and transport times for chest pain and ST-segment elevation myocardial infarction patients. *J Am Coll Cardiol*. 2012;60:806–811.
6. O'Gara PT, Kushner FG, Ascheim DD, et al. 2013 ACCF/AHA guideline for the management of ST-elevation myocardial infarction: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2013;127:e362–e425.
7. Amsterdam EA, Wenger NK, Brindis RG, et al. 2014 AHA/ACC guideline for the management of patients with non-ST-elevation acute coronary syndromes: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2014;130:e344–e426.
8. Neumann JT, Twerenbold R, Ojeda F, et al. Application of high-sensitivity troponin in suspected myocardial infarction. *N Engl J Med*. 2019;380:2529–2540.
9. Thygesen K, Alpert JS, Jaffe AS, et al. Fourth universal definition of myocardial infarction (2018). *Circulation*. 2018;138:e618–e651.
10. Becker L, Larsen MP, Eisenberg MS. Incidence of cardiac arrest during self-transport for chest pain. *Ann Emerg Med*. 1996;28:612–616.
11. Hsiao CJ, Cherry DK, Beatty PC, et al. National ambulatory medical care survey: 2007 summary. *Natl Health Stat Report*. 2010:1–32.

2.3.2. Electrocardiogram

1. Amsterdam EA, Wenger NK, Brindis RG, et al. 2014 AHA/ACC guideline for the management of patients with non-ST-elevation acute coronary syndromes: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2014;130:e344–e426.
2. O'Gara PT, Kushner FG, Ascheim DD, et al. 2013 ACCF/AHA guideline for the management of STElevation myocardial infarction: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2013;127:e362–e425.
3. Vogiatzis I, Koulouris E, Ioannidis A, et al. The importance of the 15-lead versus 12-lead ECG recordings in the diagnosis and treatment of right ventricle and left ventricle posterior and lateral wall acute myocardial infarctions. *Acta Inform Med*. 2019;27:35–39.
4. Ashida T, Tani S, Nagao K, et al. Usefulness of synthesized 18-lead electrocardiography in the diagnosis of ST-elevation myocardial infarction: a pilot study. *Am J Emerg Med*. 2017;35:448–457.
5. Matetzky S, Freimark D, Feinberg MS, et al. Acute myocardial infarction with isolated ST-segment elevation in posterior chest leads V7-9: "hidden" ST-segment elevations revealing acute posterior infarction. *J Am Coll Cardiol*. 1999;34:748–753.
6. Jain S, Ting HT, Bell M, et al. Utility of left bundle branch block as a diagnostic criterion for acute myocardial infarction. *Am J Cardiol*. 2011;107:1111–1116.
7. Ohlsson M, Ohlin H, Wallerstedt SM, et al. Usefulness of serial electrocardiograms for diagnosis of acute myocardial infarction. *Am J Cardiol*. 2001;88:478–481.
8. Mant J, McManus RJ, Oakes RA, et al. Systematic review and modelling of the investigation of acute and chronic chest pain presenting in primary care. *Health Technol Assess*. 2004;8(iii):1–158.
9. Chase M, Brown AM, Robey JL, et al. Prognostic value of symptoms during a normal or nonspecific electrocardiogram in emergency department patients with potential acute coronary syndrome. *Acad Emerg Med*. 2006;13:1034–1039.
10. Turnipseed SD, Trythall WS, Diercks DB, et al. Frequency of acute coronary syndrome in patients with normal electrocardiogram performed during presence or absence of chest pain. *Acad Emerg Med*. 2009;16:495–499.
11. Pasceri V, Cianflone D, Finocchiaro ML, et al. Relation between myocardial infarction site and pain location in Q-wave acute myocardial infarction. *Am J Cardiol*. 1995;75:224–227.
12. Pope JH, Aufderheide TP, Ruthazer R, et al. Missed diagnoses of acute cardiac ischemia in the emergency department. *N Engl J Med*. 2000;342:1163–1170.
13. McCarthy BD, Beshansky JR, D'Agostino RB, et al. Missed diagnoses of acute myocardial infarction in the emergency department: results from a multicenter study. *Ann Emerg Med*. 1993;22:579–582.
14. Canto JG, Fincher C, Kiefe CI, et al. Atypical presentations among Medicare beneficiaries with unstable angina pectoris. *Am J Cardiol*. 2002;90:248–253.
15. Savonitto S, Ardissino D, Granger CB, et al. Prognostic value of the admission electrocardiogram in acute coronary syndromes. *JAMA*. 1999;281:707–713.
16. Rouan GW, Lee TH, Cook EF, et al. Clinical characteristics and outcome of acute myocardial infarction in patients with initially normal or nonspecific electrocardiograms (a report from the Multicenter Chest Pain Study). *Am J Cardiol*. 1989;64:1087–1092.
17. Slater DK, Hlatky MA, Mark DB, et al. Outcome in suspected acute myocardial infarction with normal or minimally abnormal admission electrocardiographic findings. *Am J Cardiol*. 1987;60:766–770.
18. Fesmire FM. Which chest pain patients potentially benefit from continuous 12-lead ST-segment monitoring with automated serial ECG? *Am J Emerg Med*. 2000;18:773–778.
19. Fesmire FM, Percy RF, Bardoner JB, et al. Usefulness of automated serial 12-lead ECG monitoring during the initial emergency department evaluation of patients with chest pain. *Ann Emerg Med*. 1998;31:3–11.
20. Challa PK, Smith KM, Conti CR. Initial presenting electrocardiogram as determinant for hospital admission in patients presenting to the emergency department with chest pain: a pilot investigation. *Clin Cardiol*. 2007;30:558–561.
21. Thygesen K, Alpert JS, Jaffe AS, et al. Fourth universal definition of myocardial infarction (2018). *Circulation*. 2018;138:e618–e651.
22. Levine GN, Bates ER, Blankenship JC, et al. 2015 ACC/AHA/SCAI focused update on primary percutaneous coronary intervention for patients with ST-elevation myocardial infarction: an update of the 2011 ACCF/AHA/SCAI guideline for percutaneous coronary intervention and the 2013 ACCF/AHA guideline for the management of ST-elevation myocardial infarction. *Circulation*. 2016;133:1135–1147.

2.3.3. Chest Radiography

1. Hess EP, Perry JJ, Ladouceur P, et al. Derivation of a clinical decision rule for chest radiography in emergency department patients with chest pain and possible acute coronary syndrome. *CJEM*. 2010;12:128–134.
2. Amsterdam EA, Wenger NK, Brindis RG, et al. 2014 AHA/ACC guideline for the management of patients with non-ST-elevation acute coronary syndromes: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2014;130:e344–e426.
3. Yancy CW, Jessup M, Bozkurt B, et al. 2013 ACCF/AHA guideline for the management of heart failure: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2013;128:e240–e327.
4. Harris KM, Strauss CE, Eagle KA, et al. Correlates of delayed recognition and treatment of acute type A aortic dissection: the International Registry of Acute Aortic Dissection (IRAD). *Circulation*. 2011;124:1911–1918.

2.3.4. Biomarkers

1. Apple FS, Jesse RL, Newby LK, et al. National academy of clinical biochemistry and IFCC committee for standardization of markers of cardiac damage laboratory medicine practice guidelines: analytical issues for biochemical markers of acute coronary syndromes. *Circulation*. 2007;115:e352–e355.
2. Bandstein N, Ljung R, Johansson M, et al. Undetectable high-sensitivity cardiac troponin T level in the emergency department and risk of myocardial infarction. *J Am Coll Cardiol*. 2014;63:2569–2578.
3. Body R, Burrows G, Carley S, et al. High-sensitivity cardiac troponin t concentrations below the limit of detection to exclude acute myocardial infarction: a prospective evaluation. *Clin Chem*. 2015;61:983–989.
4. Body R, Carley S, McDowell G, et al. Rapid exclusion of acute myocardial infarction in patients with undetectable troponin using a high-sensitivity assay. *J Am Coll Cardiol*. 2011;58:1332–1339.
5. Boeddinghaus J, Nestelberger T, Twerenbold R, et al. Direct comparison of 4 very early rule-out strategies for acute myocardial infarction using high-sensitivity cardiac troponin I. *Circulation*. 2017;135:1597–1611.
6. Chapman AR, Anand A, Boeddinghaus J, et al. Comparison of the efficacy and safety of early rule-out pathways for acute myocardial infarction. *Circulation*. 2017;135:1586–1596.
7. de Lemos JA, Morrow DA, deFilippi CR. Highly sensitive troponin assays and the cardiology community: a love/hate relationship? *Clin Chem*. 2011;57:826–829.
8. Jaffe AS, Apple FS, Morrow DA, et al. Being rational about (im)precision: a statement from the Biochemistry Subcommittee of the Joint European So-

- society of Cardiology/American College of Cardiology Foundation/American Heart Association/World Heart Federation Task Force for the Definition of Myocardial Infarction. *Clin Chem*. 2010;56:941–943.
9. Morrow DA, Cannon CP, Jesse RL, et al. National academy of clinical biochemistry laboratory medicine practice guidelines: clinical characteristics and utilization of biochemical markers in acute coronary syndromes. *Circulation*. 2007;115:e356–e375.
 10. Mueller C, Giannitsis E, Christ M, et al. Multicenter evaluation of a 0-hour/1-hour algorithm in the diagnosis of myocardial infarction with high-sensitivity cardiac troponin T. *Ann Emerg Med*. 2016;68:76–87, e4.
 11. Odqvist M, Andersson PO, Tygesen H, et al. High-sensitivity troponins and outcomes after myocardial infarction. *J Am Coll Cardiol*. 2018;71:2616–2624.
 12. Peacock WF, Baumann BM, Bruton D, et al. Efficacy of high-sensitivity troponin T in identifying very-low-risk patients with possible acute coronary syndrome. *JAMA Cardiol*. 2018;3:104–111.
 13. Reichlin T, Schindler C, Drexler B, et al. One-hour rule-out and rule-in of acute myocardial infarction using high-sensitivity cardiac troponin T. *Arch Intern Med*. 2012;172:1211–1218.
 14. Reichlin T, Twerenbold R, Maushart C, et al. Risk stratification in patients with unstable angina using absolute serial changes of 3 high-sensitive troponin assays. *Am Heart J*. 2013;165:371–378.e3.
 15. Rubini Gimenez M, Hoeller R, Reichlin T, et al. Rapid rule out of acute myocardial infarction using undetectable levels of high-sensitivity cardiac troponin. *Int J Cardiol*. 2013;168:3896–3901.
 16. Rubini Gimenez M, Twerenbold R, Jaeger C, et al. One-hour rule-in and rule-out of acute myocardial infarction using high-sensitivity cardiac troponin I. *Am J Med*. 2015;128:861–870, e4.
 17. Twerenbold R, Boeddinghaus J, Nestelberger T, et al. Clinical use of high-sensitivity cardiac troponin in patients with suspected myocardial infarction. *J Am Coll Cardiol*. 2017;70:996–1012.
 18. Twerenbold R, Neumann JT, Sorensen NA, et al. Prospective validation of the 0/1-h algorithm for early diagnosis of myocardial infarction. *J Am Coll Cardiol*. 2018;72:620–632.
 19. Wildi K, Nelles B, Twerenbold R, et al. Safety and efficacy of the 0 h/3 h protocol for rapid rule out of myocardial infarction. *Am Heart J*. 2016;181:16–25.
 20. Zhelev Z, Hyde C, Youngman E, et al. Diagnostic accuracy of single baseline measurement of Elecsys Troponin T high-sensitive assay for diagnosis of acute myocardial infarction in emergency department: systematic review and meta-analysis. *BMJ*. 2015;350:h15.
 21. Thygesen K, Alpert JS, Jaffe AS, et al. Fourth universal definition of myocardial infarction (2018). *Circulation*. 2018;138:e618–e651.
 22. Cullen L, Mueller C, Parsonage WA, et al. Validation of high-sensitivity troponin I in a 2-hour diagnostic strategy to assess 30-day outcomes in emergency department patients with possible acute coronary syndrome. *J Am Coll Cardiol*. 2013;62:1242–1249.
 23. Lipinski MJ, Baker NC, Escarcega RO, et al. Comparison of conventional and high-sensitivity troponin in patients with chest pain: a collaborative meta-analysis. *Am Heart J*. 2015;169:6–16, e6.
 24. Twerenbold R, Wildi K, Jaeger C, et al. Optimal cutoff levels of more sensitive cardiac troponin assays for the early diagnosis of myocardial infarction in patients with renal dysfunction. *Circulation*. 2015;131:2041–2050.
 25. van Wijk S, Jacobs L, Eurlings LW, et al. Troponin T measurements by high-sensitivity vs conventional assays for risk stratification in acute dyspnea. *Clin Chem*. 2012;58:284–292.
 26. Neumann JT, Twerenbold R, Ojeda F, et al. Application of high-sensitivity troponin in suspected myocardial infarction. *N Engl J Med*. 2019;380:2529–2540.
 27. Aviles RJ, Wright RS, Aviles JM, et al. Long-term prognosis of patients with clinical unstable angina pectoris without elevation of creatine kinase but with elevation of cardiac troponin I levels. *Am J Cardiol*. 2002;90:875–878.
 28. Eggers KM, Oldgren J, Nordenskjold A, et al. Diagnostic value of serial measurement of cardiac markers in patients with chest pain: limited value of adding myoglobin to troponin I for exclusion of myocardial infarction. *Am Heart J*. 2004;148:574–581.
 29. Kavsak PA, MacRae AR, Newman AM, et al. Effects of contemporary troponin assay sensitivity on the utility of the early markers myoglobin and CKMB isoforms in evaluating patients with possible acute myocardial infarction. *Clin Chim Acta*. 2007;380:213–216.
 30. Kontos MC, de Lemos JA, Ou FS, et al. Troponin-positive, MB-negative patients with non-ST-elevation myocardial infarction: an undertreated but high-risk patient group: results from the National Cardiovascular Data Registry Acute Coronary Treatment and Intervention Outcomes Network-Get With The Guidelines (NCDR ACTION-GWTG) Registry. *Am Heart J*. 2010;160:819–825.
 31. Newby LK, Roe MT, Chen AY, et al. Frequency and clinical implications of discordant creatine kinase-MB and troponin measurements in acute coronary syndromes. *J Am Coll Cardiol*. 2006;47:312–318.
 32. Volz KA, McGillicuddy DC, Horowitz GL, et al. Creatine kinase-MB does not add additional benefit to a negative troponin in the evaluation of chest pain. *Am J Emerg Med*. 2012;30:188–190.
 33. Reichlin T, Twerenbold R, Wildi K, et al. Prospective validation of a 1-hour algorithm to rule-out and rule-in acute myocardial infarction using a high-sensitivity cardiac troponin T assay. *CMAJ*. 2015;187:E243–E252.
 34. Shah AS, Griffiths M, Lee KK, et al. High sensitivity cardiac troponin and the under-diagnosis of myocardial infarction in women: prospective cohort study. *BMJ*. 2015;350:g7873.
 35. Morrow DA. The Fourth Universal Definition of Myocardial Infarction and the Emerging Importance of Myocardial Injury. *Circulation*. 2020;141:172–175.

3. CARDIAC TESTING GENERAL CONSIDERATIONS

1. Wolk MJ, Bailey SR, Doherty JU, et al. ACCF/AHA/ASE/ASNC/HFSA/HRS/SCAI/SCCT/SCMR/STS 2013 multimodality appropriate use criteria for the detection and risk assessment of stable ischemic heart disease: a report of the American College of Cardiology Foundation Appropriate Use Criteria Task Force, American Heart Association, American Society of Echocardiography, American Society of Nuclear Cardiology, Heart Failure Society of America, Heart Rhythm Society, Society for Cardiovascular Angiography and Interventions, Society of Cardiovascular Computed Tomography, Society for Cardiovascular Magnetic Resonance, and Society of Thoracic Surgeons. *J Am Coll Cardiol*. 2014;63:380–406.
2. Medicare Program: Changes to Hospital Outpatient Prospective Payment and Ambulatory Surgical Center Payment Systems and Quality Reporting Programs; Revisions of Organ Procurement Organizations Conditions of Coverage; Prior Authorization Process and Requirements for Certain Covered Outpatient Department Services; Potential Changes to the Laboratory Date of Service Policy; Changes to Grandfathered Children's Hospitals-Within-Hospitals; Notice of Closure of Two Teaching Hospitals and Opportunity to Apply for Available Slots. Available at: <https://www.cms.gov/Medicare/Medicare-Fee-for-Service-Payment/HospitalOutpatientPPS/Hospital-Outpatient-Regulations-and-Notices-Items/CMS-1717-FC>. Accessed September 14, 2020.

3.1. Anatomic Testing

3.1.1. Coronary Computed Tomography Angiography

1. Gaur S, Ovrehus KA, Dey D, et al. Coronary plaque quantification and fractional flow reserve by coronary computed tomography angiography identify ischaemia-causing lesions. *Eur Heart J*. 2016;37:1220–1227.
2. Ferencik M, Mayrhofer T, Bittner DO, et al. Use of high-risk coronary atherosclerotic plaque detection for risk stratification of patients with stable chest pain: a secondary analysis of the PROMISE randomized clinical trial. *JAMA Cardiol*. 2018;3:144–152.
3. Budoff MJ, Mayrhofer T, Ferencik M, et al. Prognostic value of coronary artery calcium in the PROMISE Study (Prospective Multicenter Imaging Study for Evaluation of Chest Pain). *Circulation*. 2017;136:1993–2005.
4. Ferencik M, Mayrhofer T, Puchner SB, et al. Computed tomography-based high-risk coronary plaque score to predict acute coronary syndrome among patients with acute chest pain—Results from the ROMICAT II trial. *J Cardiovasc Comput Tomogr*. 2015;9:538–545.
5. Puchner SB, Liu T, Mayrhofer T, et al. High-risk plaque detected on coronary CT angiography predicts acute coronary syndromes independent of significant stenosis in acute chest pain: results from the ROMICAT-II trial. *J Am Coll Cardiol*. 2014;64:684–692.
6. Motoyama S, Ito H, Sarai M, et al. Plaque characterization by coronary computed tomography angiography and the likelihood of acute coronary events in mid-term follow-up. *J Am Coll Cardiol*. 2015;66:337–346.

- Einstein AJ, Berman DS, Min JK, et al. Patient-centered imaging: shared decision making for cardiac imaging procedures with exposure to ionizing radiation. *J Am Coll Cardiol*. 2014;63:1480–1489.
- Einstein AJ. Effects of radiation exposure from cardiac imaging: how good are the data? *J Am Coll Cardiol*. 2012;59:553–565.
- Fazel R, Gerber TC, Balter S, et al. Approaches to enhancing radiation safety in cardiovascular imaging: a scientific statement from the American Heart Association. *Circulation*. 2014;130:1730–1748.

3.2.4. Cardiovascular Magnetic Resonance Imaging

- Henzlova MJ, Duvall WL, Einstein AJ, et al. ASNC imaging guidelines for SPECT nuclear cardiology procedures: stress, protocols, and tracers. *J Nucl Cardiol*. 2016;23:606–639.
- Senior R, Becher H, Monaghan M, et al. Contrast echocardiography: evidence-based recommendations by European Association of Echocardiography. *Eur J Echocardiogr*. 2009;10:194–212.
- Patil HR, Main M. Revisiting the safety profile of echocardiography contrast agents. Available at: <https://www.acc.org/latest-in-cardiology/articles/2016/06/23/08/23/revisiting-the-safety-profile-of-echocardiography-contrast-agents>. Accessed August 16, 2020.
- Porter TR, Abdelmoneim S, Belcik JT, et al. Guidelines for the cardiac sonographer in the performance of contrast echocardiography: a focused update from the American Society of Echocardiography. *J Am Soc Echocardiogr*. 2014;27:797–810.
- Pellikka PA, Arruda-Olson A, Chaudhry FA, et al. Guidelines for performance, interpretation, and application of stress echocardiography in ischemic heart disease: from the American Society of Echocardiography. *J Am Soc Echocardiogr*. 2020;33:1–41.e8.
- Kramer CM, Barkhausen J, Bucciarelli-Ducci C, et al. Standardized cardiovascular magnetic resonance imaging (CMR) protocols: 2020 update. *J Cardiovasc Magn Reson*. 2020;22:17.
- Narula J, Chandrasekhar Y, Ahmadi A, et al. SCCT 2021 Expert consensus document on coronary computed tomographic angiography: a report of the Society of Cardiovascular Computed Tomography. *J Cardiovasc Comput Tomogr*. 2021;15:192–217.
- Gibbons RJ, Balady GJ, Beasley JW, et al. ACC/AHA Guidelines for exercise testing. a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines (Committee on Exercise Testing). *J Am Coll Cardiol*. 1997;30:260–311.
- Fletcher GF, Ades PA, Kligfield P, et al. Exercise standards for testing and training: a scientific statement from the American Heart Association. *Circulation*. 2013;128:873–934.
- Henzlova MJ, Cerqueira MD, Mahmarian JJ, et al. Stress protocols and tracers. *J Nucl Cardiol*. 2006;13:e80–e90.
- SNM Procedure Guideline for General Imaging 6.0 Available at: https://s3.amazonaws.com/rdcms-snm/1/files/production/public/docs/General_Imaging_Version_6.0.pdf. Accessed July 26, 2021.
- Committee on Obstetric Practice. Committee Opinion No. 723: Guidelines for diagnostic imaging during pregnancy and lactation. *Obstet Gynecol*. 2017;130:e210–e216.
- Ashwath M. Safety of CMR During Pregnancy and Lactation. Available at: <https://scmr.org/page/Pregnancy>. Accessed November 9, 2020.
- ACR-SPR practice parameter for imaging pregnant or potentially pregnant adolescents and women with ionizing radiation. Available at: <https://www.acr.org/-/media/ACR/Files/Practice-Parameters/pregnant-pts.pdf>. Accessed November 9, 2020.

3.3. Cardiac Testing Considerations for Women Who Are Pregnant, Postpartum, or of Child-Bearing Age

- Image Wisely. Available at: <https://imagewisely.org>. Accessed September 10, 2020.
- Committee on Obstetric Practice. Committee Opinion No. 723: Guidelines for diagnostic imaging during pregnancy and lactation. *Obstet Gynecol*. 2017;130:e210–e216.
- Ray JG, Vermeulen MJ, Bharatha A, et al. Association between MRI exposure during pregnancy and fetal and childhood outcomes. *JAMA*. 2016;316:952–961.
- Kodwara R. ACR manual on contrast media: 2018 updates. *Radiol Technol*. 2019;91:97–100.

- Expert Panel on MRScanal E, Barkovich AJ, et al. ACR guidance document on MR safe practices: 2013 *J Magn Reson Imaging*. 2013;37:501–530.
- Truong QA, Rinehart S, Abbata S, et al. Coronary computed tomographic imaging in women: an expert consensus statement from the Society of Cardiovascular Computed Tomography. *J Cardiovasc Comput Tomogr*. 2018;12:451–466.

4. CHOOSING THE RIGHT PATHWAY WITH PATIENT-CENTRIC ALGORITHMS FOR ACUTE CHEST PAIN

4.1. Patients With Acute Chest Pain and Suspected ACS (Not Including STEMI)

- Apple FS, Jesse RL, Newby LK, et al. National academy of clinical biochemistry and IFCC committee for standardization of markers of cardiac damage laboratory medicine practice guidelines: analytical issues for biochemical markers of acute coronary syndromes. *Circulation*. 2007;115:e352–e355.
- Bandstein N, Jung R, Johansson M, et al. Undetectable high-sensitivity cardiac troponin T level in the emergency department and risk of myocardial infarction. *J Am Coll Cardiol*. 2014;63:2569–2578.
- Body R, Burrows G, Carley S, et al. High-sensitivity cardiac troponin t concentrations below the limit of detection to exclude acute myocardial infarction: a prospective evaluation. *Clin Chem*. 2015;61:983–989.
- Body R, Carley S, McDowell G, et al. Rapid exclusion of acute myocardial infarction in patients with undetectable troponin using a high-sensitivity assay. *J Am Coll Cardiol*. 2011;58:1332–1339.
- Boeddinghaus J, Nestelberger T, Twerenbold R, et al. Direct comparison of 4 very early rule-out strategies for acute myocardial infarction using high-sensitivity cardiac troponin I. *Circulation*. 2017;135:1597–1611.
- Chapman AR, Anand A, Boeddinghaus J, et al. Comparison of the efficacy and safety of early rule-out pathways for acute myocardial infarction. *Circulation*. 2017;135:1586–1596.
- de Lemos JA, Morrow DA, deFilippi CR. Highly sensitive troponin assays and the cardiology community: a love/hate relationship? *Clin Chem*. 2011;57:826–829.
- Jaffe AS. Chasing troponin: how low can you go if you can see the rise? *J Am Coll Cardiol*. 2006;48:1763–1764.
- Jaffe AS, Apple FS, Morrow DA, et al. Being rational about (im)precision: a statement from the Biochemistry Subcommittee of the Joint European Society of Cardiology/American College of Cardiology Foundation/American Heart Association/World Heart Federation Task Force for the Definition of Myocardial Infarction. *Clin Chem*. 2010;56:941–943.
- Morrow DA, Cannon CP, Jesse RL, et al. National Academy of Clinical Biochemistry laboratory medicine practice guidelines: clinical characteristics and utilization of biochemical markers in acute coronary syndromes. *Circulation*. 2007;115:e356–e375.
- Mueller C, Giannitsis E, Christ M, et al. Multicenter evaluation of a 0-hour/1-hour algorithm in the diagnosis of myocardial infarction with high-sensitivity cardiac troponin T. *Ann Emerg Med*. 2016;68:76–87.e4.
- Odqvist M, Andersson PO, Tygesen H, et al. High-sensitivity troponins and outcomes after myocardial infarction. *J Am Coll Cardiol*. 2018;71:2616–2624.
- Peacock WF, Baumann BM, Bruton D, et al. Efficacy of high-sensitivity troponin T in identifying very-low-risk patients with possible acute coronary syndrome. *JAMA Cardiol*. 2018;3:104–111.
- Reichlin T, Schindler C, Drexler B, et al. One-hour rule-out and rule-in of acute myocardial infarction using high-sensitivity cardiac troponin T. *Arch Intern Med*. 2012;172:1211–1218.
- Morrow DA. Clinician's guide to early rule-out strategies with high-sensitivity cardiac troponin. *Circulation*. 2017;135:1612–1616.
- Thygesen K, Alpert JS, Jaffe AS, et al. Fourth universal definition of myocardial infarction (2018). *Circulation*. 2018;138:e618–e651.
- Twerenbold R, Boeddinghaus J, Nestelberger T, et al. Clinical use of high-sensitivity cardiac troponin in patients with suspected myocardial infarction. *J Am Coll Cardiol*. 2017;70:996–1012.
- Januzzi JL Jr, Mahler SA, Christenson RH, et al. Recommendations for institutions transitioning to high-sensitivity troponin testing: JACC Scientific Expert Panel. *J Am Coll Cardiol*. 2019;73:1059–1077.
- Huis In 't Veld MA, Cullen L, Mahler SA, et al. The fast and the furious: low-risk chest pain and the rapid rule-out protocol. *West J Emerg Med*. 2017;18:474–478.

20. Schulman-Marcus J, Harteigh BO, Gransar H, et al. Sex-specific associations between coronary artery plaque extent and risk of major adverse cardiovascular events: the CONFIRM Long-Term Registry. *J Am Coll Cardiol Img.* 2016;9:364–372.
21. Hadamitzky M, Freissmuth B, Meyer T, et al. Prognostic value of coronary computed tomographic angiography for prediction of cardiac events in patients with suspected coronary artery disease. *J Am Coll Cardiol Img.* 2009;2:404–411.
22. Finck T, Hardenberg J, Will A, et al. Ten-year follow-up after coronary computed tomography angiography in patients with suspected coronary artery disease. *J Am Coll Cardiol Img.* 2019;12:1330–1338.
23. Hochman JS, Reynolds HR, Bangalore S, et al. Baseline characteristics and risk profiles of participants in the ISCHEMIA randomized clinical trial. *JAMA Cardiol.* 2019;4:273–286.
24. ISCHEMIA Trial Research Group, Maron DJ, Hochman JS, et al. International Study of Comparative Health Effectiveness with Medical and Invasive Approaches (ISCHEMIA) trial: rationale and design. *Am Heart J.* 2018;201:124–135.
25. Pickering JW, Than MP, Cullen L, et al. Rapid rule-out of acute myocardial infarction with a single high-sensitivity cardiac troponin T measurement below the limit of detection: a collaborative meta-analysis. *Ann Intern Med.* 2017;166:715–724.
26. McRae AD, Innes G, Graham M, et al. Undetectable concentrations of a Food and Drug Administration-approved high-sensitivity cardiac troponin T assay to rule out acute myocardial infarction at emergency department arrival. *Acad Emerg Med.* 2017;24:1267–1277.
27. Sandoval Y, Nowak R, deFilippi CR, et al. Myocardial infarction risk stratification with a single measurement of high-sensitivity troponin I. *J Am Coll Cardiol.* 2019;74:271–282.
28. Bularga A, Lee KK, Stewart S, et al. High-sensitivity troponin and the application of risk stratification thresholds in patients with suspected acute coronary syndrome. *Circulation.* 2019;140:1557–1568.
29. Chew DP, Lambrakis K, Blyth A, et al. A randomized trial of a 1-hour troponin T protocol in suspected acute coronary syndromes: the Rapid Assessment of Possible Acute Coronary Syndrome in the emergency department with high-sensitivity Troponin T Study (RAPID-TnT). *Circulation.* 2019;140:1543–1556.
30. Than M, Cullen L, Aldous S, et al. 2-Hour accelerated diagnostic protocol to assess patients with chest pain symptoms using contemporary troponins as the only biomarker: the ADAPT trial. *J Am Coll Cardiol.* 2012;59:2091–2098.
31. Mahler SA, Riley RF, Hiestand BC, et al. The HEART Pathway randomized trial: identifying emergency department patients with acute chest pain for early discharge. *Circ Cardiovasc Qual Outcomes.* 2015;8:195–203.
32. Mark DG, Huang J, Chettipally U, et al. Performance of coronary risk scores among patients with chest pain in the emergency department. *J Am Coll Cardiol.* 2018;71:606–616.
33. Stoppyra JP, Miller CD, Hiestand BC, et al. Chest pain risk stratification: a comparison of the 2-Hour Accelerated Diagnostic Protocol (ADAPT) and the HEART pathway. *Crit Pathw Cardiol.* 2016;15:46–49.
34. Stoppyra JP, Miller CD, Hiestand BC, et al. Validation of the no objective testing rule and comparison to the HEART Pathway. *Acad Emerg Med.* 2017;24:1165–1168.
35. Stoppyra JP, Riley RF, Hiestand BC, et al. The HEART Pathway randomized controlled trial one-year outcomes. *Acad Emerg Med.* 2019;26:41–50.
36. Keller T, Zeller T, Peetz D, et al. Sensitive troponin I assay in early diagnosis of acute myocardial infarction. *N Engl J Med.* 2009;361:868–877.
37. Neumann JT, Twerenbold R, Ojeda F, et al. Application of high-sensitivity troponin in suspected myocardial infarction. *N Engl J Med.* 2019;380:2529–2540.
38. Roffi M, Patrono C, Collet JP, et al. 2015 ESC guidelines for the management of acute coronary syndromes in patients presenting without persistent ST-segment elevation: Task Force for the Management of Acute Coronary Syndromes in Patients Presenting Without Persistent ST-Segment Elevation of the European Society of Cardiology (ESC). *Eur Heart J.* 2016;37:267–315.
39. Thygesen K, Mair J, Giannitsis E, et al. How to use high-sensitivity cardiac troponins in acute cardiac care. *Eur Heart J.* 2012;33:2252–2257.
40. Cullen L, Mueller C, Parsonage WA, et al. Validation of high-sensitivity troponin I in a 2-hour diagnostic strategy to assess 30-day outcomes in emergency department patients with possible acute coronary syndrome. *J Am Coll Cardiol.* 2013;62:1242–1249.
41. Reichlin T, Twerenbold R, Wildi K, et al. Prospective validation of a 1-hour algorithm to rule-out and rule-in acute myocardial infarction using a high-sensitivity cardiac troponin T assay. *CMAJ.* 2015;187:E243–E252.
42. Rubini Gimenez M, Twerenbold R, Jaeger C, et al. One-hour rule-in and rule-out of acute myocardial infarction using high-sensitivity cardiac troponin I. *Am J Med.* 2015;128:861–870, e4.
43. Twerenbold R, Neumann JT, Sorensen NA, et al. Prospective validation of the 0/1-h algorithm for early diagnosis of myocardial infarction. *J Am Coll Cardiol.* 2018;72:620–632.
44. Than MP, Pickering JW, Aldous SJ, et al. Effectiveness of EDACS versus ADAPT accelerated diagnostic pathways for chest pain: a pragmatic randomized controlled trial embedded within practice. *Ann Emerg Med.* 2016;68:93–102 e1.
45. Than M, Aldous S, Lord SJ, et al. A 2-hour diagnostic protocol for possible cardiac chest pain in the emergency department: a randomized clinical trial. *JAMA Intern Med.* 2014;174:51–58.
46. Collet JP, Thiele H, Barbato E, et al. 2020 ESC guidelines for the management of acute coronary syndromes in patients presenting without persistent ST-segment elevation. *Eur Heart J.* 2021;42:1289–1367.
47. Twerenbold R, Costabel JP, Nestelberger T, et al. Outcome of applying the ESC 0/1-hour algorithm in patients with suspected myocardial infarction. *J Am Coll Cardiol.* 2019;74:483–494.
48. Backus BE, Six AJ, Kelder JC, et al. A prospective validation of the HEART score for chest pain patients at the emergency department. *Int J Cardiol.* 2013;168:2153–2158.
49. Fanaroff AC, Rymer JA, Goldstein SA, et al. Does this patient with chest pain have acute coronary syndrome?: the rational clinical examination systematic review. *JAMA.* 2015;314:1955–1965.
50. McCord J, Cabrera R, Lindahl B, et al. Prognostic utility of a modified HEART score in chest pain patients in the emergency department. *Circ Cardiovasc Qual Outcomes.* 2017;10:e003101.
51. Bank IEM, de Hoog VC, de Kleijn DPV, et al. Sex-based differences in the performance of the HEART score in patients presenting to the emergency department with acute chest pain. *J Am Heart Assoc.* 2017;6:e005373.
52. Romero-Farina G, Candell-Riera J, Aguade-Bruix S, et al. Warranty periods for normal myocardial perfusion stress SPECT. *J Nucl Cardiol.* 2015;22:44–54.
53. Twerenbold R, Wildi K, Jaeger C, et al. Optimal cutoff levels of more sensitive cardiac troponin assays for the early diagnosis of myocardial infarction in patients with renal dysfunction. *Circulation.* 2015;131:2041–2050.
54. Wildi K, Nelles B, Twerenbold R, et al. Safety and efficacy of the 0 h/3 h protocol for rapid rule out of myocardial infarction. *Am Heart J.* 2016;181:16–25.
55. Zhelev Z, Hyde C, Youngman E, et al. Diagnostic accuracy of single baseline measurement of Elecsys Troponin T high-sensitive assay for diagnosis of acute myocardial infarction in emergency department: systematic review and meta-analysis. *BMJ.* 2015;350:h15.

4.1.1. Low Risk Patients With Acute Chest Pain

1. Bandstein N, Ljung R, Johansson M, et al. Undetectable high-sensitivity cardiac troponin T level in the emergency department and risk of myocardial infarction. *J Am Coll Cardiol.* 2014;63:2569–2578.
2. Chapman AR, Anand A, Boeddinghaus J, et al. Comparison of the efficacy and safety of early rule-out pathways for acute myocardial infarction. *Circulation.* 2017;135:1586–1596.
3. Mueller C, Giannitsis E, Christ M, et al. Multicenter evaluation of a 0-hour/1-hour algorithm in the diagnosis of myocardial infarction with high-sensitivity cardiac troponin T. *Ann Emerg Med.* 2016;68:76–87, e4.
4. Reichlin T, Schindler C, Drexler B, et al. One-hour rule-out and rule-in of acute myocardial infarction using high-sensitivity cardiac troponin T. *Arch Intern Med.* 2012;172:1211–1218.
5. Reichlin T, Twerenbold R, Wildi K, et al. Prospective validation of a 1-hour algorithm to rule-out and rule-in acute myocardial infarction using a high-sensitivity cardiac troponin T assay. *CMAJ.* 2015;187:E243–E252.
6. Rubini Gimenez M, Hoeller R, Reichlin T, et al. Rapid rule out of acute myocardial infarction using undetectable levels of high-sensitivity cardiac troponin. *Int J Cardiol.* 2013;168:3896–3901.
7. Rubini Gimenez M, Twerenbold R, Jaeger C, et al. One-hour rule-in and rule-out of acute myocardial infarction using high-sensitivity cardiac troponin I. *Am J Med.* 2015;128:861–870, e4.
8. Twerenbold R, Neumann JT, Sorensen NA, et al. Prospective validation of the 0/1-h algorithm for early diagnosis of myocardial infarction. *J Am Coll Cardiol.* 2018;72:620–632.
9. Pickering JW, Than MP, Cullen L, et al. Rapid rule-out of acute myocardial infarction with a single high-sensitivity cardiac troponin T measurement be-

low the limit of detection: a collaborative meta-analysis. *Ann Intern Med*. 2017;166:715–724.

10. Peacock WF, Baumann BM, Bruton D, et al. Efficacy of high-sensitivity troponin T in identifying very-low-risk patients with possible acute coronary syndrome. *JAMA Cardiol*. 2018;3:104–111.
11. Neumann JT, Twerenbold R, Ojeda F, et al. Application of high-sensitivity troponin in suspected myocardial infarction. *N Engl J Med*. 2019;380:2529–2540.
12. Greenslade JH, Carlton EW, Van Hise C, et al. Diagnostic accuracy of a new high-sensitivity troponin I assay and five accelerated diagnostic pathways for ruling out acute myocardial infarction and acute coronary syndrome. *Ann Emerg Med*. 2018;71:439–451, e3.
13. Cullen L, Mueller C, Parsonage WA, et al. Validation of high-sensitivity troponin I in a 2-hour diagnostic strategy to assess 30-day outcomes in emergency department patients with possible acute coronary syndrome. *J Am Coll Cardiol*. 2013;62:1242–1249.
14. Flaws D, Than M, Scheuermeyer FX, et al. External validation of the emergency department assessment of chest pain score accelerated diagnostic pathway (EDACS-ADP). *Emerg Med J*. 2016;33:618–625.
15. Stopyna JP, Miller CD, Hiestand BC, et al. Validation of the no objective testing rule and comparison to the HEART Pathway. *Acad Emerg Med*. 2017;24:1165–1168.
16. Than M, Aldous S, Lord SJ, et al. A 2-hour diagnostic protocol for possible cardiac chest pain in the emergency department: a randomized clinical trial. *JAMA Intern Med*. 2014;174:51–58.
17. Twerenbold R, Boeddinghaus J, Nestelberger T, et al. Clinical use of high-sensitivity cardiac troponin in patients with suspected myocardial infarction. *J Am Coll Cardiol*. 2017;70:996–1012.
18. Mahler SA, Lenoir KM, Wells BJ, et al. Safely identifying emergency department patients with acute chest pain for early discharge. *Circulation*. 2018;138:2456–2468.
19. Amsterdam EA, Wenger NK, Brindis RG, et al. 2014 AHA/ACC guideline for the management of patients with non-ST-elevation acute coronary syndromes: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2014;130:e344–e426.
20. Mahler SA, Riley RF, Hiestand BC, et al. The HEART Pathway randomized trial: identifying emergency department patients with acute chest pain for early discharge. *Circ Cardiovasc Qual Outcomes*. 2015;8:195–203.
21. Than M, Cullen L, Aldous S, et al. 2-Hour accelerated diagnostic protocol to assess patients with chest pain symptoms using contemporary troponins as the only biomarker: the ADAPT trial. *J Am Coll Cardiol*. 2012;59:2091–2098.

4.1.1.1. Cost-Value Considerations in the Evaluation of Low-Risk Patients

1. Velickovic VM, Rochau U, Conrads-Frank A, et al. Systematic assessment of decision-analytic models evaluating diagnostic tests for acute myocardial infarction based on cardiac troponin assays. *Expert Rev Pharmacoecon Outcomes Res*. 2018;1–22.
2. Westwood M, van Asselt T, Ramaekers B, et al. High-sensitivity troponin assays for the early rule-out or diagnosis of acute myocardial infarction in people with acute chest pain: a systematic review and cost-effectiveness analysis. *Health Technol Assess*. 2015;19:1–234.
3. Goodacre S, Thokala P, Carroll C, et al. Systematic review, meta-analysis and economic modelling of diagnostic strategies for suspected acute coronary syndrome. *Health Technol Assess*. 2013;17:v–vi, 1–188.
4. Polanczyk CA, Kuntz KM, Sacks DB, et al. Emergency department triage strategies for acute chest pain using creatine kinase-MB and troponin I assays: a cost-effectiveness analysis. *Ann Intern Med*. 1999;131:909–918.
5. Poldervaart JM, Reitsma JB, Backus BE, et al. Effect of using the HEART score in patients with chest pain in the emergency department: a stepped-wedge, cluster randomized trial. *Ann Intern Med*. 2017;166:689–697.
6. Riley RF, Miller CD, Russell GB, et al. Cost analysis of the History, ECG, Age, Risk factors, and initial Troponin (HEART) Pathway randomized control trial. *Am J Emerg Med*. 2017;35:77–81.
7. Parsonage WA, Milburn T, Ashover S, et al. Implementing change: evaluating the Accelerated Chest Pain Risk Evaluation (ACRE) project. *Med J Aust*. 2017;207:201–205.
8. Julicher P, Greenslade JH, Parsonage WA, et al. The organisational value of diagnostic strategies using high-sensitivity troponin for patients with possible acute coronary syndromes: a trial-based cost-effectiveness analysis. *BMJ Open*. 2017;7:e013653.

9. Cheng Q, Greenslade JH, Parsonage WA, et al. Change to costs and lengths of stay in the emergency department and the Brisbane protocol: an observational study. *BMJ Open*. 2016;6:e009746.
10. Roberts RR, Zalenski RJ, Mensah EK, et al. Costs of an emergency department-based accelerated diagnostic protocol vs hospitalization in patients with chest pain: a randomized controlled trial. *JAMA*. 1997;278:1670–1676.
11. Kip MMA, Koffijberg H, Moesker MJ, et al. The cost-utility of point-of-care troponin testing to diagnose acute coronary syndrome in primary care. *BMC Cardiovasc Disord*. 2017;17:213.
12. Vaidya A, Severens JL, Bongaerts BW, et al. High-sensitive troponin T assay for the diagnosis of acute myocardial infarction: an economic evaluation. *BMC Cardiovasc Disord*. 2014;14:77.
13. Twerenbold R, Jaeger C, Rubini Gimenez M, et al. Impact of high-sensitivity cardiac troponin on use of coronary angiography, cardiac stress testing, and time to discharge in suspected acute myocardial infarction. *Eur Heart J*. 2016;37:3324–3332.

4.1.2. Intermediate-Risk Patients With Acute Chest Pain

1. Farkouh ME, Smars PA, Reeder GS, et al. A clinical trial of a chest-pain observation unit for patients with unstable angina. Chest Pain Evaluation in the Emergency Room (CHEER) Investigators. *N Engl J Med*. 1998;339:1882–1888.
2. Miller CD, Hwang W, Hoekstra JW, et al. Stress cardiac magnetic resonance imaging with observation unit care reduces cost for patients with emergent chest pain: a randomized trial. *Ann Emerg Med*. 2010;201056:209–219.
3. Roberts RR, Zalenski RJ, Mensah EK, et al. Costs of an emergency department-based accelerated diagnostic protocol vs hospitalization in patients with chest pain: a randomized controlled trial. *JAMA*. 1997;278:1670–1676.
4. Ross MA, Hockenberry JM, Mutter R, et al. Protocol-driven emergency department observation units offer savings, shorter stays, and reduced admissions. *Health Aff (Millwood)*. 2013;32:2149–2156.
5. Rydman RJ, Zalenski RJ, Roberts RR, et al. Patient satisfaction with an emergency department chest pain observation unit. *Ann Emerg Med*. 1997;29:109–115.
6. Miller CD, Hwang W, Case D, et al. Stress CMR imaging observation unit in the emergency department reduces 1-year medical care costs in patients with acute chest pain: a randomized study for comparison with inpatient care. *J Am Coll Cardiol Img*. 2011;4:862–870.
7. Hockenberry JM, Mutter R, Barrett M, et al. Factors associated with prolonged observation services stays and the impact of long stays on patient cost. *Health Serv Res*. 2014;49:893–909.
8. Labovitz AJ, Noble VE, Bierig M, et al. Focused cardiac ultrasound in the emergent setting: a consensus statement of the American Society of Echocardiography and American College of Emergency Physicians. *J Am Soc Echocardiogr*. 2010;23:1225–1230.
9. Roffi M, Patrono C, Collet JP, et al. 2015 ESC guidelines for the management of acute coronary syndromes in patients presenting without persistent ST-segment elevation: Task Force for the Management of Acute Coronary Syndromes in Patients Presenting Without Persistent ST-Segment Elevation of the European Society of Cardiology (ESC). *Eur Heart J*. 2016;37:267–315.
10. Amsterdam EA, Wenger NK, Brindis RG, et al. 2014 AHA/ACC guideline for the management of patients with non-ST-elevation acute coronary syndromes: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2014;130:e344–e426.
11. Douglas PS, Garcia MJ, Haines D, et al. ACCF/ASE/AHA/ASNC/HFSA/HRS/SCAI/SCCM/SCCT/SCMR 2011 appropriate use criteria for echocardiography: a report of the American College of Cardiology Foundation Appropriate Use Criteria Task Force, American Society of Echocardiography, American Heart Association, American Society of Nuclear Cardiology, Heart Failure Society of America, Heart Rhythm Society, Society for Cardiovascular Angiography and Interventions, Society of Critical Care Medicine, Society of Cardiovascular Computed Tomography, and Society for Cardiovascular Magnetic Resonance. *J Am Coll Cardiol*. 2011;57:1126–1166.
12. Sicari R, Nihoyannopoulos P, Evangelista A, et al. Stress echocardiography expert consensus statement—executive summary: European Association of Echocardiography (EAE) (a registered branch of the ESC). *Eur Heart J*. 2009;30:278–289.
13. Health Quality O. Stress echocardiography for the diagnosis of coronary artery disease: an evidence-based analysis. *Ont Health Technol Assess Ser*. 2010;10:1–61.

4.1.2.1. Intermediate-Risk Patients With Acute Chest Pain and No Known CAD

- Goldstein JA, Chinnaiyan KM, Abidov A, et al. The CT-STAT (Coronary Computed Tomographic Angiography for Systematic Triage of Acute Chest Pain Patients to Treatment) trial. *J Am Coll Cardiol*. 2011;58:1414–1422.
- Litt HI, Gatsonis C, Snyder B, et al. CT angiography for safe discharge of patients with possible acute coronary syndromes. *N Engl J Med*. 2012;366:1393–1403.
- Goodacre S, Thokala P, Carroll C, et al. Systematic review, meta-analysis and economic modelling of diagnostic strategies for suspected acute coronary syndrome. *Health Technol Assess*. 2013;17:v–vi; 1–188.
- Goldstein JA, Gallagher MJ, O'Neill WW, et al. A randomized controlled trial of multi-slice coronary computed tomography for evaluation of acute chest pain. *J Am Coll Cardiol*. 2007;49:863–871.
- Chang SA, Choi SI, Choi EK, et al. Usefulness of 64-slice multidetector computed tomography as an initial diagnostic approach in patients with acute chest pain. *Am Heart J*. 2008;156:375–383.
- Miller AH, Pepe PE, Peshock R, et al. Is coronary computed tomography angiography a resource sparing strategy in the risk stratification and evaluation of acute chest pain? Results of a randomized controlled trial. *Acad Emerg Med*. 2011;18:458–467.
- Hoffmann U, Truong QA, Schoenfeld DA, et al. Coronary CT angiography versus standard evaluation in acute chest pain. *N Engl J Med*. 2012;367:299–308.
- Linde JJ, Kofoed KF, Sogaard M, et al. Cardiac computed tomography guided treatment strategy in patients with recent acute-onset chest pain: results from the randomised, controlled trial: CARDiac CT in the treatment of acute CHest pain (CATCH) *Int J Cardiol*. 2013;168:5257–5262.
- Hamilton-Craig C, Fifoot A, Hansen M, et al. Diagnostic performance and cost of CT angiography versus stress ECG—a randomized prospective study of suspected acute coronary syndrome chest pain in the emergency department (CT-COMPARE) *Int J Cardiol*. 2014;177:867–873.
- Levsky JM, Spevack DM, Travin MI, et al. Coronary computed tomography angiography versus radionuclide myocardial perfusion imaging in patients with chest pain admitted to telemetry: a randomized trial. *Ann Intern Med*. 2015;163:174–183.
- Dedic A, Lubbers MM, Schaap J, et al. Coronary CT angiography for suspected ACS in the era of high-sensitivity troponins: randomized multicenter study. *J Am Coll Cardiol*. 2016;67:16–26.
- Hoffmann U, Ferencik M, Udelsom JE, et al. Prognostic value of noninvasive cardiovascular testing in patients with stable chest pain: insights from the PROMISE trial (Prospective Multicenter Imaging Study for Evaluation of Chest Pain) *Circulation*. 2017;135:2320–2332.
- Min JK, Dunning A, Lin FY, et al. Age- and sex-related differences in all-cause mortality risk based on coronary computed tomography angiography findings results from the International Multicenter CONFIRM (Coronary CT Angiography Evaluation for Clinical Outcomes: An International Multicenter Registry) of 23 854 patients without known coronary artery disease. *J Am Coll Cardiol*. 2011;58:849–860.
- Blankstein R, Ahmed W, Bamberg F, et al. Comparison of exercise treadmill testing with cardiac computed tomography angiography among patients presenting to the emergency room with chest pain: the Rule Out Myocardial Infarction Using Computer-Assisted Tomography (ROMICAT) study. *Circ Cardiovasc Imaging*. 2012;5:233–242.
- Jeetley P, Burden L, Stoykova B, et al. Clinical and economic impact of stress echocardiography compared with exercise electrocardiography in patients with suspected acute coronary syndrome but negative troponin: a prospective randomized controlled study. *Eur Heart J*. 2007;28:204–211.
- Dadkhah S, Almuwaqqat Z, Sulaiman S, et al. Sensitive troponin I and stress testing in the emergency department for the early management of chest pain using 2-hour protocol. *Crit Pathw Cardiol*. 2017;16:89–92.
- Nucifora G, Badano LP, Sarraf-Zadegan N, et al. Comparison of early dobutamine stress echocardiography and exercise electrocardiographic testing for management of patients presenting to the emergency department with chest pain. *Am J Cardiol*. 2007;100:1068–1073.
- Jasani G, Papas M, Patel AJ, et al. Immediate stress echocardiography for low-risk chest pain patients in the emergency department: a prospective observational cohort study. *J Emerg Med*. 2018;54:156–164.
- Krishnan S, Venn R, Blumenthal DM, et al. Utilization of stress testing for low-risk patients with chest discomfort in the emergency department. *J Nucl Cardiol*. 2019;26:1642–1646.
- Hermann LK, Newman DH, Pleasant WA, et al. Yield of routine provocative cardiac testing among patients in an emergency department-based chest pain unit. *JAMA Intern Med*. 2013;173:1128–1133.
- Diercks DB, Mumma BE, Frank Peacock W, et al. Incremental value of objective cardiac testing in addition to physician impression and serial contemporary troponin measurements in women. *Acad Emerg Med*. 2013;20:265–270.
- Poldervaart JM, Six AJ, Backus BE, et al. The predictive value of the exercise ECG for major adverse cardiac events in patients who presented with chest pain in the emergency department. *Clin Res Cardiol*. 2013;102:305–312.
- Napoli AM, Tran S, Wang J. Low-risk chest pain patients younger than 40 years do not benefit from admission and stress testing. *Crit Pathw Cardiol*. 2013;12:201–203.
- Scott AC, Bilesky J, Lamanna A, et al. Limited utility of exercise stress testing in the evaluation of suspected acute coronary syndrome in patients aged less than 40 years with intermediate risk features. *Emerg Med Australas*. 2014;26:170–176.
- Aldous S, Richards AM, Cullen L, et al. The incremental value of stress testing in patients with acute chest pain beyond serial cardiac troponin testing. *Emerg Med J*. 2016;33:319–324.
- Greenslade JH, Parsonage W, Ho A, et al. Utility of routine exercise stress testing among intermediate risk chest pain patients attending an emergency department. *Heart Lung Circ*. 2015;24:879–884.
- Skoien W. Diagnostic Yield of Routine Stress Testing in low and intermediate risk chest pain patients under 40 years: a systematic review. *Crit Pathw Cardiol*. 2016;15:114–120.
- Amsterdam EA, Wenger NK, Brindis RG, et al. 2014 AHA/ACC guideline for the management of patients with non-ST-elevation acute coronary syndromes: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2014;130:e344–e426.
- Levsky JM, Haramati LB, Spevack DM, et al. Coronary computed tomography angiography versus stress echocardiography in acute chest pain: a randomized controlled trial. *J Am Coll Cardiol*. 2018;11:1288–1297.
- Heitner JF, Klem I, Rasheed D, et al. Stress cardiac MR imaging compared with stress echocardiography in the early evaluation of patients who present to the emergency department with intermediate-risk chest pain. *Radiology*. 2014;271:56–64.
- Udelsom JE, Beshansky JR, Ballin DS, et al. Myocardial perfusion imaging for evaluation and triage of patients with suspected acute cardiac ischemia: a randomized controlled trial. *JAMA*. 2002;288:2693–2700.
- Miller CD, Case LD, Little WC, et al. Stress CMR reduces revascularization, hospital readmission, and recurrent cardiac testing in intermediate-risk patients with acute chest pain. *J Am Coll Cardiol*. 2013;61:785–794.
- Miller CD, Hwang W, Case D, et al. Stress CMR imaging observation unit in the emergency department reduces 1-year medical care costs in patients with acute chest pain: a randomized study for comparison with inpatient care. *J Am Coll Cardiol*. 2011;4:862–870.
- Miller CD, Hwang W, Hoekstra JW, et al. Stress cardiac magnetic resonance imaging with observation unit care reduces cost for patients with emergent chest pain: a randomized trial. *Ann Emerg Med*. 2010;20:2056:209–219.
- Ingkanisorn WP, Kwong RY, Bohme NS, et al. Prognosis of negative adenosine stress magnetic resonance in patients presenting to an emergency department with chest pain. *J Am Coll Cardiol*. 2006;47:1427–1432.
- Siontis GC, Mavridis D, Greenwood JP, et al. Outcomes of non-invasive diagnostic modalities for the detection of coronary artery disease: network meta-analysis of diagnostic randomised controlled trials. *BMJ*. 2018;360:k504.
- Norgaard BL, Leipsic J, Gaur S, et al. Diagnostic performance of noninvasive fractional flow reserve derived from coronary computed tomography angiography in suspected coronary artery disease: the NXT trial (Analysis of Coronary Blood Flow Using CT Angiography: Next Steps) *J Am Coll Cardiol*. 2014;63:1145–1155.
- Sand NPR, Veien KT, Nielsen SS, et al. Prospective comparison of FFR derived from coronary CT angiography with SPECT perfusion imaging in stable coronary artery disease: the ReASSESS study. *J Am Coll Cardiol*. 2018;11:1640–1650.
- Patel MR, Norgaard BL, Fairbairn TA, et al. 1-Year impact on medical practice and clinical outcomes of FFRCT: the ADVANCE registry. *J Am Coll Cardiol*. 2020;13:97–105.
- Fairbairn TA, Nieman K, Akasaka T, et al. Real-world clinical utility and impact on clinical decision-making of coronary computed tomography angiography-derived fractional flow reserve: lessons from the ADVANCE Registry. *Eur Heart J*. 2018;39:3701–3711.
- Nakanishi R, Osawa K, Ceponiene I, et al. The diagnostic performance of SPECT-MPI to predict functional significant coronary artery disease by fractional flow reserve derived from CCTA (FFRCT): sub-analysis from ACCURACY and VCT001 studies. *Int J Cardiovasc Imaging*. 2017;33:2067–2072.

42. Chinnaiyan KM, Safian RD, Gallagher ML, et al. Clinical use of CT-derived fractional flow reserve in the emergency department. *J Am Coll Cardiol Img.* 2020;13:452–461.
43. Ferencik M, Lu MT, Mayrhofer T, et al. Non-invasive fractional flow reserve derived from coronary computed tomography angiography in patients with acute chest pain: subgroup analysis of the ROMICAT II trial. *J Cardiovasc Comput Tomogr.* 2019;13:196–202.
44. Schulman-Marcus J, Hartaigh BO, Gransar H, et al. Sex-specific associations between coronary artery plaque extent and risk of major adverse cardiovascular events: the CONFIRM long-term registry. *J Am Coll Cardiol Img.* 2016;9:364–372.
45. Hadamitzky M, Freissmuth B, Meyer T, et al. Prognostic value of coronary computed tomographic angiography for prediction of cardiac events in patients with suspected coronary artery disease. *J Am Coll Cardiol Img.* 2009;2:404–411.
46. Finck T, Hardenberg J, Will A, et al. Ten-year follow-up after coronary computed tomography angiography in patients with suspected coronary artery disease. *J Am Coll Cardiol Img.* 2019;12:1330–1338.
47. Hulten E, Pickett C, Bittencourt MS, et al. Outcomes after coronary computed tomography angiography in the emergency department: a systematic review and meta-analysis of randomized, controlled trials. *J Am Coll Cardiol.* 2013;61:880–892.
48. Linde JJ, Hove JD, Sogaard M, et al. Long-term clinical impact of coronary CT angiography in patients with recent acute-onset chest pain: the randomized controlled CATCH trial. *J Am Coll Cardiol Img.* 2015;8:1404–1413.
49. Hulten E, Pickett C, Bittencourt MS, et al. Meta-analysis of coronary CT angiography in the emergency department. *Eur Heart J Cardiovasc Imaging.* 2013;14:607.
50. D'Ascenzo F, Cerrato E, Biondi-Zoccai G, et al. Coronary computed tomographic angiography for detection of coronary artery disease in patients presenting to the emergency department with chest pain: a meta-analysis of randomized clinical trials. *Eur Heart J Cardiovasc Imaging.* 2013;14:782–789.
51. Hachamovitch R, Nutter B, Hlatky MA, et al. Patient management after non-invasive cardiac imaging results from SPARC (Study of myocardial perfusion and coronary anatomy imaging roles in coronary artery disease) *J Am Coll Cardiol.* 2012;59:462–474.
52. Shaw LJ, Berman DS, Picard MH, et al. Comparative definitions for moderate-severe ischemia in stress nuclear, echocardiography, and magnetic resonance imaging. *J Am Coll Cardiol Img.* 2014;7:593–604.
53. Fihn SD, Gardin JM, Abrams J, et al. 2012 ACCF/AHA/ACP/AATS/PCNA/SCAI/STS guideline for the diagnosis and management of patients with stable ischemic heart disease: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines, and the American College of Physicians, American Association for Thoracic Surgery, Preventive Cardiovascular Nurses Association, Society for Cardiovascular Angiography and Interventions, and Society of Thoracic Surgeons. *Circulation.* 2012;126:e354–e471.
54. Roffi M, Patrono C, Collet JP, et al. 2015 ESC Guidelines for the management of acute coronary syndromes in patients presenting without persistent ST-segment elevation: Task Force for the Management of Acute Coronary Syndromes in Patients Presenting without Persistent ST-Segment Elevation of the European Society of Cardiology (ESC) *Eur Heart J.* 2016;37:267–315.
55. Khan A, Engineer R, Wang S, et al. Initial experience regarding the safety and yield of rest-stress myocardial perfusion imaging in emergency department patients with mildly abnormal high-sensitivity cardiac troponins. *J Nucl Cardiol.* 2020;10.1007/s12350-020-02145-w.
56. Litt HI, Gatsonis C, Snyder B, et al. CT angiography for safe discharge of patients with possible acute coronary syndromes. *N Engl J Med.* 2012;366:1393–1403.
57. Hoffmann U, Truong QA, Schoenfeld DA, et al. Coronary CT angiography versus standard evaluation in acute chest pain. *N Engl J Med.* 2012;367:299–308.
58. Hulten E, Pickett C, Bittencourt MS, et al. Outcomes after coronary computed tomography angiography in the emergency department: a systematic review and meta-analysis of randomized, controlled trials. *J Am Coll Cardiol.* 2013;61:880–892.
59. D'Ascenzo F, Cerrato E, Biondi-Zoccai G, et al. Coronary computed tomographic angiography for detection of coronary artery disease in patients presenting to the emergency department with chest pain: a meta-analysis of randomized clinical trials. *Eur Heart J Cardiovasc Imaging.* 2013;14:782–789.
60. Boden WE, O'Rourke RA, Teo KK, et al. Optimal medical therapy with or without PCI for stable coronary disease. *N Engl J Med.* 2007;356:1503–1516.
61. Maron DJ, Hochman JS, Reynolds HR, et al. Initial invasive or conservative strategy for stable coronary disease. *N Engl J Med.* 2020;382:1395–1407.
62. Hoffmann U, Truong QA, Schoenfeld DA, et al. Coronary CT angiography versus standard evaluation in acute chest pain. *N Engl J Med.* 2012;367:299–308.
63. Goldstein JA, Chinnaiyan KM, Abidov A, et al. The CT-STAT (Coronary Computed Tomographic Angiography for Systematic Triage of Acute Chest Pain Patients to Treatment) trial. *J Am Coll Cardiol.* 2011;58:1414–1422.
64. Litt HI, Gatsonis C, Snyder B, et al. CT angiography for safe discharge of patients with possible acute coronary syndromes. *N Engl J Med.* 2012;366:1393–1403.
65. Hulten E, Pickett C, Bittencourt MS, et al. Outcomes after coronary computed tomography angiography in the emergency department: a systematic review and meta-analysis of randomized, controlled trials. *J Am Coll Cardiol.* 2013;61:880–892.
66. Goodacre S, Thokala P, Carroll C, et al. Systematic review, meta-analysis and economic modelling of diagnostic strategies for suspected acute coronary syndrome. *Health Technol Assess.* 2013;17:v–vi; 1–188.
67. Hulten E, Pickett C, Bittencourt MS, et al. Meta-analysis of coronary CT angiography in the emergency department. *Eur Heart J Cardiovasc Imaging.* 2013;14:607.
68. Puchner SB, Liu T, Mayrhofer T, et al. High-risk plaque detected on coronary CT angiography predicts acute coronary syndromes independent of significant stenosis in acute chest pain: results from the ROMICAT-II trial. *J Am Coll Cardiol.* 2014;64:684–692.
69. Motoyama S, Ito H, Sarai M, et al. Plaque characterization by coronary computed tomography angiography and the likelihood of acute coronary events in mid-term follow-up. *J Am Coll Cardiol.* 2015;66:337–346.
70. Chang HJ, Lin FY, Lee SE, et al. Coronary atherosclerotic precursors of acute coronary syndromes. *J Am Coll Cardiol.* 2018;71:2511–2522.
71. Douglas PS, Pontone G, Hlatky MA, et al. Clinical outcomes of fractional flow reserve by computed tomographic angiography-guided diagnostic strategies vs. usual care in patients with suspected coronary artery disease: the prospective longitudinal trial of FFR(CT): outcome and resource impacts study. *Eur Heart J.* 2015;36:3359–3367.
72. Norgaard BL, Leipsic J, Gaur S, et al. Diagnostic performance of noninvasive fractional flow reserve derived from coronary computed tomography angiography in suspected coronary artery disease: the NXT trial (Analysis of Coronary Blood Flow Using CT Angiography: Next Steps). *J Am Coll Cardiol.* 2014;63:1145–1155.
73. Patel MR, Norgaard BL, Fairbairn TA, et al. 1-Year impact on medical practice and clinical outcomes of FFRCT: the ADVANCE registry. *J Am Coll Cardiol Img.* 2020;13:97–105.
74. Fairbairn TA, Nieman K, Akasaka T, et al. Real-world clinical utility and impact on clinical decision-making of coronary computed tomography angiography-derived fractional flow reserve: lessons from the ADVANCE Registry. *Eur Heart J.* 2018;39:3701–3711.
75. Chinnaiyan KM, Safian RD, Gallagher ML, et al. Clinical use of CT-derived fractional flow reserve in the emergency department. *J Am Coll Cardiol Img.* 2020;13:452–461.
76. Ferencik M, Lu MT, Mayrhofer T, et al. Non-invasive fractional flow reserve derived from coronary computed tomography angiography in patients with

4.1.2.1.1. Cost-Value Considerations

1. Jasani G, Papas M, Patel AJ, et al. Immediate stress echocardiography for low-risk chest pain patients in the emergency department: a prospective observational cohort study. *J Emerg Med.* 2018;54:156–164.
2. Davies R, Liu G, Sciamanna C, et al. Comparison of the effectiveness of stress echocardiography versus myocardial perfusion imaging in patients presenting to the emergency department with low-risk chest pain. *Am J Cardiol.* 2016;118:1786–1791.
3. Levsky JM, Haramati LB, Spevack DM, et al. Coronary computed tomography angiography versus stress echocardiography in acute chest pain: a randomized controlled trial. *J Am Coll Cardiol Img.* 2018;11:1288–1297.
4. Goldstein JA, Chinnaiyan KM, Abidov A, et al. The CT-STAT (Coronary Computed Tomographic Angiography for Systematic Triage of Acute Chest Pain Patients to Treatment) trial. *J Am Coll Cardiol.* 2011;58:1414–1422.

4.1.2.2. Intermediate-Risk Patients With Acute Chest Pain and Known CAD

1. Boden WE, O'Rourke RA, Teo KK, et al. Optimal medical therapy with or without PCI for stable coronary disease. *N Engl J Med.* 2007;356:1503–1516.
2. Maron DJ, Hochman JS, Reynolds HR, et al. Initial invasive or conservative strategy for stable coronary disease. *N Engl J Med.* 2020;382:1395–1407.
3. Hoffmann U, Truong QA, Schoenfeld DA, et al. Coronary CT angiography versus standard evaluation in acute chest pain. *N Engl J Med.* 2012;367:299–308.
4. Goldstein JA, Chinnaiyan KM, Abidov A, et al. The CT-STAT (Coronary Computed Tomographic Angiography for Systematic Triage of Acute Chest Pain Patients to Treatment) trial. *J Am Coll Cardiol.* 2011;58:1414–1422.
5. Litt HI, Gatsonis C, Snyder B, et al. CT angiography for safe discharge of patients with possible acute coronary syndromes. *N Engl J Med.* 2012;366:1393–1403.
6. Hulten E, Pickett C, Bittencourt MS, et al. Outcomes after coronary computed tomography angiography in the emergency department: a systematic review and meta-analysis of randomized, controlled trials. *J Am Coll Cardiol.* 2013;61:880–892.
7. Goodacre S, Thokala P, Carroll C, et al. Systematic review, meta-analysis and economic modelling of diagnostic strategies for suspected acute coronary syndrome. *Health Technol Assess.* 2013;17:v–vi; 1–188.
8. Hulten E, Pickett C, Bittencourt MS, et al. Meta-analysis of coronary CT angiography in the emergency department. *Eur Heart J Cardiovasc Imaging.* 2013;14:607.
9. Puchner SB, Liu T, Mayrhofer T, et al. High-risk plaque detected on coronary CT angiography predicts acute coronary syndromes independent of significant stenosis in acute chest pain: results from the ROMICAT-II trial. *J Am Coll Cardiol.* 2014;64:684–692.
10. Motoyama S, Ito H, Sarai M, et al. Plaque characterization by coronary computed tomography angiography and the likelihood of acute coronary events in mid-term follow-up. *J Am Coll Cardiol.* 2015;66:337–346.
11. Chang HJ, Lin FY, Lee SE, et al. Coronary atherosclerotic precursors of acute coronary syndromes. *J Am Coll Cardiol.* 2018;71:2511–2522.
12. Douglas PS, Pontone G, Hlatky MA, et al. Clinical outcomes of fractional flow reserve by computed tomographic angiography-guided diagnostic strategies vs. usual care in patients with suspected coronary artery disease: the prospective longitudinal trial of FFR(CT): outcome and resource impacts study. *Eur Heart J.* 2015;36:3359–3367.
13. Norgaard BL, Leipsic J, Gaur S, et al. Diagnostic performance of noninvasive fractional flow reserve derived from coronary computed tomography angiography in suspected coronary artery disease: the NXT trial (Analysis of Coronary Blood Flow Using CT Angiography: Next Steps). *J Am Coll Cardiol.* 2014;63:1145–1155.
14. Patel MR, Norgaard BL, Fairbairn TA, et al. 1-Year impact on medical practice and clinical outcomes of FFRCT: the ADVANCE registry. *J Am Coll Cardiol Img.* 2020;13:97–105.
15. Fairbairn TA, Nieman K, Akasaka T, et al. Real-world clinical utility and impact on clinical decision-making of coronary computed tomography angiography-derived fractional flow reserve: lessons from the ADVANCE Registry. *Eur Heart J.* 2018;39:3701–3711.
16. Chinnaiyan KM, Safian RD, Gallagher ML, et al. Clinical use of CT-derived fractional flow reserve in the emergency department. *J Am Coll Cardiol Img.* 2020;13:452–461.
17. Ferencik M, Lu MT, Mayrhofer T, et al. Non-invasive fractional flow reserve derived from coronary computed tomography angiography in patients with

acute chest pain: subgroup analysis of the ROMICAT II trial. *J Cardiovasc Comput Tomogr*. 2019;13:196–202.

18. Udelson JE, Beshansky JR, Ballin DS, et al. Myocardial perfusion imaging for evaluation and triage of patients with suspected acute cardiac ischemia: a randomized controlled trial. *JAMA*. 2002;288:2693–2700.
19. Nucifora G, Badano LP, Sarraf-Zadegan N, et al. Comparison of early dobutamine stress echocardiography and exercise electrocardiographic testing for management of patients presenting to the emergency department with chest pain. *Am J Cardiol*. 2007;100:1068–1073.
20. Linde JJ, Kofoed KF, Sogaard M, et al. Cardiac computed tomography guided treatment strategy in patients with recent acute-onset chest pain: results from the randomised, controlled trial: CARDiac cT in the treatment of acute CHEst pain (CATCH) *Int J Cardiol*. 2013;168:5257–5262.
21. Blankstein R, Ahmed W, Bamberg F, et al. Comparison of exercise treadmill testing with cardiac computed tomography angiography among patients presenting to the emergency room with chest pain: the Rule Out Myocardial Infarction Using Computer-Assisted Tomography (ROMICAT) study. *Circ Cardiovasc Imaging*. 2012;5:233–242.
22. Shaw LJ, Berman DS, Maron DJ, et al. Optimal medical therapy with or without percutaneous coronary intervention to reduce ischemic burden: results from the Clinical Outcomes Utilizing Revascularization and Aggressive Drug Evaluation (COURAGE) trial nuclear substudy. *Circulation*. 2008;117:1283–1291.
23. Boden NPR, O'Rourke RA, Crawford MH, et al. Outcomes in patients with acute non-Q-wave myocardial infarction randomly assigned to an invasive as compared with a conservative management strategy. Veterans Affairs Non-Q-Wave Infarction Strategies in Hospital (VANQWISH) Trial Investigators. *N Engl J Med*. 1998;338:1785–1792.
24. Lee SE, Chang HJ, Sung JM, et al. Effects of statins on coronary atherosclerotic plaques: the PARADIGM study. *J Am Coll Cardiol Img*. 2018;11:1475–1484.
25. Sand NPR, Veien KT, Nielsen SS, et al. Prospective comparison of FFR derived from coronary ct angiography with SPECT perfusion imaging in stable coronary artery disease: the ReASSESS study. *J Am Coll Cardiol Img*. 2018;11:1640–1650.
26. Fihn SD, Gardin JM, Abrams J, et al. 2012 ACCF/AHA/ACP/AATS/PCNA/SCAI/STS guideline for the diagnosis and management of patients with stable ischemic heart disease: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines, and the American College of Physicians, American Association for Thoracic Surgery, Preventive Cardiovascular Nurses Association, Society for Cardiovascular Angiography and Interventions, and Society of Thoracic Surgeons. *Circulation*. 2012;126:e354–e471.

4.1.3. High-Risk Patients With Acute Chest Pain

1. Mahmarian JJ, Shaw LJ, Filipchuk NG, et al. A multinational study to establish the value of early adenosine technetium-99m sestamibi myocardial perfusion imaging in identifying a low-risk group for early hospital discharge after acute myocardial infarction. *J Am Coll Cardiol*. 2006;48:2448–2457.
2. Mahmarian JJ, Dakik HA, Filipchuk NG, et al. An initial strategy of intensive medical therapy is comparable to that of coronary revascularization for suppression of scintigraphic ischemia in high-risk but stable survivors of acute myocardial infarction. *J Am Coll Cardiol*. 2006;48:2458–2467.
3. Yan AT, Yan RT, Tan M, et al. Risk scores for risk stratification in acute coronary syndromes: useful but simpler is not necessarily better. *Eur Heart J*. 2007;28:1072–1078.
4. Amsterdam EA, Wenger NK, Brindis RG, et al. 2014 AHA/ACC guideline for the management of patients with non-ST-elevation acute coronary syndromes: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2014;130:e344–e426.
5. Patel MR, Calhoun JH, Dehmer GJ, et al. ACC/AATS/AHA/ASE/ASNC/SCAI/SCCT/STS 2017 appropriate use criteria for coronary revascularization in patients with stable ischemic heart disease: a report of the American College of Cardiology Appropriate Use Criteria Task Force, American Association for Thoracic Surgery, American Heart Association, American Society of Echocardiography, American Society of Nuclear Cardiology, Society for Cardiovascular Angiography and Interventions, Society of Cardiovascular Computed Tomography, and Society of Thoracic Surgeons. *J Am Coll Cardiol*. 2017;69:2212–2241.
6. Patel MR, Bailey SR, Bonow RO, et al. ACCF/SCAI/AATS/AHA/ASE/ASNC/HFSA/HRS/SCCM/SCCT/SCMR/STS 2012 appropriate use criteria for diagnostic catheterization: a report of the American College of Cardiology Foundation Appropriate Use Criteria Task Force, Society for

Cardiovascular Angiography and Interventions, American Association for Thoracic Surgery, American Heart Association, American Society of Echocardiography, American Society of Nuclear Cardiology, Heart Failure Society of America, Heart Rhythm Society, Society of Critical Care Medicine, Society of Cardiovascular Computed Tomography, Society for Cardiovascular Magnetic Resonance, and Society of Thoracic Surgeons. *J Am Coll Cardiol*. 2012;59:1995–2027.

7. Levine GN, Bates ER, Blankenship JC, et al. 2015 ACC/AHA/SCAI focused update on primary percutaneous coronary intervention for patients with STElevation myocardial infarction: an update of the 2011 ACCF/AHA/SCAI guideline for percutaneous coronary intervention and the 2013 ACCF/AHA guideline for the management of ST-Elevation myocardial infarction. *Circulation*. 2016;133:1135–1147.
8. Pathik B, Raman B, Mohd Amin NH, et al. Troponin-positive chest pain with unobstructed coronary arteries: incremental diagnostic value of cardiovascular magnetic resonance imaging. *Eur Heart J Cardiovasc Imaging*. 2016;17:1146–1152.
9. Dastidar AG, Rodrigues JCL, Johnson TW, et al. Myocardial infarction with nonobstructed coronary arteries: impact of CMR early after presentation. *J Am Coll Cardiol Img*. 2017;10:1204–1206.
10. Assomull RG, Lyne JC, Keenan N, et al. The role of cardiovascular magnetic resonance in patients presenting with chest pain, raised troponin, and unobstructed coronary arteries. *Eur Heart J*. 2007;28:1242–1249.
11. Tornvall P, Gerbaud E, Behaghel A, et al. Myocarditis or "true" infarction by cardiac magnetic resonance in patients with a clinical diagnosis of myocardial infarction without obstructive coronary disease: a meta-analysis of individual patient data. *Atherosclerosis*. 2015;241:87–91.
12. Dastidar AG, Baritussio A, De Garate E, et al. Prognostic role of CMR and conventional risk factors in myocardial infarction with nonobstructed coronary arteries. *J Am Coll Cardiol Img*. 2019;12:1973–1982.
13. Greenslade JH, Carlton EW, Van Hise C, et al. Diagnostic accuracy of a new high-sensitivity troponin I assay and five accelerated diagnostic pathways for ruling out acute myocardial infarction and acute coronary syndrome. *Ann Emerg Med*. 2018;71:439–451, e3.
14. Raff GL, Hoffmann U, Udelson JE. Trials of imaging use in the emergency department for acute chest pain. *J Am Coll Cardiol Img*. 2017;10:338–349.
15. Cohen M. High-risk acute coronary syndrome patients with non-ST-elevation myocardial infarction: definition and treatment. *Cardiovasc Drugs Ther*. 2008;22:407–418.
16. Steg PG, Fitzgerald G, Fox KA. Risk stratification in non-ST-segment elevation acute coronary syndromes: troponin alone is not enough. *Am J Med*. 2009;122:107–108.
17. Pasupathy S, Air T, Dreyer RP, et al. Systematic review of patients presenting with suspected myocardial infarction and nonobstructive coronary arteries. *Circulation*. 2015;131:861–870.
18. Dastidar AG, Rodrigues JC, Ahmed N, et al. The role of cardiac MRI in patients with troponin-positive chest pain and unobstructed coronary arteries. *Curr Cardiovasc Imaging Rep*. 2015;8:28.
19. Niccoli G, Scalone G, Crea F. Acute myocardial infarction with no obstructive coronary atherosclerosis: mechanisms and management. *Eur Heart J*. 2015;36:475–481.

4.1.4. Acute Chest Pain in Patients With Prior Coronary Artery Bypass Graft (CABG) Surgery

1. Fitzgibbon GM, Kafka HP, Leach AJ, et al. Coronary bypass graft fate and patient outcome: angiographic follow-up of 5065 grafts related to survival and reoperation in 1388 patients during 25 years. *J Am Coll Cardiol*. 1996;28:616–626.
2. Harskamp RE, Lopes RD, Baisden CE, et al. Saphenous vein graft failure after coronary artery bypass surgery: pathophysiology, management, and future directions. *Ann Surg*. 2013;257:824–833.
3. Wolk MJ, Bailey SR, Doherty JU, et al. ACCF/AHA/ASE/ASNC/HFSA/HRS/SCAI/SCCT/SCMR/STS 2013 multimodality appropriate use criteria for the detection and risk assessment of stable ischemic heart disease: a report of the American College of Cardiology Foundation Appropriate Use Criteria Task Force, American Heart Association, American Society of Echocardiography, American Society of Nuclear Cardiology, Heart Failure Society of America, Heart Rhythm Society, Society for Cardiovascular Angiography and Interventions, Society of Cardiovascular Computed Tomography, Society for Cardiovascular Magnetic Resonance, and Society of Thoracic Surgeons. *J Am Coll Cardiol*. 2014;63:380–406.
4. Taylor AJ, Cerqueira M, Hodgson JM, et al. ACCF/SCCT/ACR/AHA/ASE/ASNC/NASCI/SCAI/SCMR 2010 appropriate use criteria for cardiac computed tomography: a report of the American College of Cardiol-

- ogy Foundation Appropriate Use Criteria Task Force, the Society of Cardiovascular Computed Tomography, the American College of Radiology, the American Heart Association, the American Society of Echocardiography, the American Society of Nuclear Cardiology, the North American Society for Cardiovascular Imaging, the Society for Cardiovascular Angiography and Interventions, and the Society for Cardiovascular Magnetic Resonance. *Circulation*. 2010;122:e525–e555.
5. Sabik JF 3rd. Understanding saphenous vein graft patency. *Circulation*. 2011;124:273–275.
 6. Taggart DP. Current status of arterial grafts for coronary artery bypass grafting. *Ann Cardiothorac Surg*. 2013;2:427–430.
 7. Gaudino M, Benedetto U, Fremes S, et al. Radial-artery or saphenous-vein grafts in coronary-artery bypass surgery. *N Engl J Med*. 2018;378:2069–2077.
 8. Fihn SD, Blankenship JC, Alexander KP, et al. 2014 ACC/AHA/AATS/PCNA/SCAI/STS focused update of the guideline for the diagnosis and management of patients with stable ischemic heart disease: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines, and the American Association for Thoracic Surgery, Preventive Cardiovascular Nurses Association, Society for Cardiovascular Angiography and Interventions, and Society of Thoracic Surgeons. *Circulation*. 2014;130:1749–1767.
 9. Hammermeister KE, DeRouen TA, Dodge HT. Variables predictive of survival in patients with coronary disease. Selection by univariate and multivariate analyses from the clinical, electrocardiographic, exercise, arteriographic, and quantitative angiographic evaluations. *Circulation*. 1979;59:421–430.
 10. Mark DB, Hlatky MA, Harrell FE Jr, et al. Exercise treadmill score for predicting prognosis in coronary artery disease. *Ann Intern Med*. 1987;106:793–800.
 11. Levine GN, Bates ER, Blankenship JC, et al. 2011 ACCF/AHA/SCAI guideline for percutaneous coronary intervention: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines and the Society for Cardiovascular Angiography and Interventions. *Circulation*. 2011;124:e574–e651.
 12. Hillis LD, Smith PK, Anderson JL, et al. 2011 ACCF/AHA guideline for coronary artery bypass graft surgery: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2011;124:e652–e735.
 13. Patel MR, Bailey SR, Bonow RO, et al. ACCF/SCAI/AATS/AHA/ASE/ASNC/HFSA/HRS/SCCM/SCCT/SCMR/STS 2012 appropriate use criteria for diagnostic catheterization: a report of the American College of Cardiology Foundation Appropriate Use Criteria Task Force, Society for Cardiovascular Angiography and Interventions, American Association for Thoracic Surgery, American Heart Association, American Society of Echocardiography, American Society of Nuclear Cardiology, Heart Failure Society of America, Heart Rhythm Society, Society of Critical Care Medicine, Society of Cardiovascular Computed Tomography, Society for Cardiovascular Magnetic Resonance, and Society of Thoracic Surgeons. *J Am Coll Cardiol*. 2012;59:1995–2027.
 14. Bittl JA, He Y, Jacobs AK, et al. Bayesian methods affirm the use of percutaneous coronary intervention to improve survival in patients with unprotected left main coronary artery disease. *Circulation*. 2013;127:2177–2185.
 15. Costa MA, Trentini CA, Schafranski MD, et al. Factors associated with the development of chronic post-sternotomy pain: a case-control study. *Braz J Cardiovasc Surg*. 2015;30:552–556.
 16. van Gulik L, Janssen LI, Ahlers SJ, et al. Risk factors for chronic thoracic pain after cardiac surgery via sternotomy. *Eur J Cardiothorac Surg*. 2011;40:1309–1313.
 17. Kalso E, Mennander S, Tasmuth T, et al. Chronic post-sternotomy pain. *Acta Anaesthesiol Scand*. 2001;45:935–939.
 18. Rashidi S, Elenbaas TW, Hamad MA, et al. Does removal of steel wires relieve post-sternotomy pain after cardiac surgery? *Asian Cardiovasc Thorac Ann*. 2013;21:409–413.
 19. Lahtinen P, Kokki H, Hynynen M. Pain after cardiac surgery: a prospective cohort study of 1-year incidence and intensity. *Anesthesiology*. 2006;105:794–800.
 20. Marcassa C, Faggiano P, Greco C, et al. A retrospective multicenter study on long-term prevalence of chronic pain after cardiac surgery. *J Cardiovasc Med (Hagerstown)*. 2015;16:768–774.
 21. Barbero U, Iannaccone M, d'Ascenzo F, et al. 64 slice-coronary computed tomography sensitivity and specificity in the evaluation of coronary artery bypass graft stenosis: a meta-analysis. *Int J Cardiol*. 2016;216:52–57.

4.1.5. Evaluation of Patients With Acute Chest Pain Receiving Dialysis

1. Saran R, Robinson B, Abbott KC, et al. US renal data system 2017 annual data report: epidemiology of kidney disease in the United States. *Am J Kidney Dis*. 2018;71:A7.
2. Voroneanu L, Covic A. Arrhythmias in hemodialysis patients. *J Nephrol*. 2009;22:716–725.
3. Herzog CA, Littrell K, Arko C, et al. Clinical characteristics of dialysis patients with acute myocardial infarction in the United States: a collaborative project of the United States Renal Data System and the National Registry of Myocardial Infarction. *Circulation*. 2007;116:1465–1472.
4. Kielstein JT, Abou-Rebyeh F, Hafer C, et al. Right-sided chest pain at the onset of haemodialysis. *Nephrol Dial Transplant*. 2001;16:1493–1495.
5. Modi KS, Gross D, Davidman M. The patient developing chest pain at the onset of haemodialysis sessions—it is not always angina pectoris. *Nephrol Dial Transplant*. 1999;14:221–223.
6. Merritt B, Naamon E, Morris SA. The influence of an Urgent Care Center on the frequency of ED visits in an urban hospital setting. *Am J Emerg Med*. 2000;18:123–125.
7. Weinick RM, Burns RM, Mehrotra A. Many emergency department visits could be managed at urgent care centers and retail clinics. *Health Aff (Millwood)*. 2010;29:1630–1636.

4.1.6. Evaluation of Acute Chest Pain in Patients With Cocaine and Methamphetamine Use

1. Finkel JB, Marhefka GD. Rethinking cocaine-associated chest pain and acute coronary syndromes. *Mayo Clin Proc*. 2011;86:1198–1207.
2. DeFilippis E, Singh A, Divakaran S, et al. Cocaine and marijuana use among young adults with myocardial infarction. *J Am Coll Cardiol*. 2018;71:2540–2551.
3. Hawley LA, Auten JD, Matteucci MJ, et al. Cardiac complications of adult methamphetamine exposures. *J Emerg Med*. 2013;45:821–827.
4. McCord J, Jneid H, Hollander JE, et al. Management of cocaine-associated chest pain and myocardial infarction: a scientific statement from the American Heart Association Acute Cardiac Care Committee of the Council on Clinical Cardiology. *Circulation*. 2008;117:1897–1907.
5. Rezkalla SH, Kloner RA. Cocaine-induced acute myocardial infarction. *Clin Med Res*. 2007;5:172–176.
6. Richards JR, Hamidi S, Grant CD, et al. Methamphetamine use and emergency department utilization: 20 Years Later. *J Addict*. 2017;2017:4050932.
7. Wei GL, Zheng XZ, Chen KQ, et al. Coronary sinus flow is reduced in methamphetamine abusers: a transthoracic echocardiographic study. *Int J Cardiovasc Imaging*. 2018;34:1889–1894.
8. Paratz ED, Zhao J, Sherwen AK, et al. Is an abnormal ECG just the tip of the ICE-berg? Examining the utility of electrocardiography in detecting methamphetamine-induced cardiac pathology. *Heart Lung Circ*. 2017;26:684–689.
9. Paratz ED, Cunningham NJ, MacIsaac AI. The cardiac complications of methamphetamines. *Heart Lung Circ*. 2016;25:325–332.

4.1.7. Shared Decision-Making in Patients With Acute Chest Pain

1. Hess EP, Hollander JE, Schaffer JT, et al. Shared decision making in patients with low risk chest pain: prospective randomized pragmatic trial. *BMJ*. 2016;355:i6165.
2. Hess EP, Knoedler MA, Shah ND, et al. The chest pain choice decision aid: a randomized trial. *Circ Cardiovasc Qual Outcomes*. 2012;5:251–259.
3. Mayo Clinic. Chest Pain Choice Decision Aid. YouTube. 2016. Available at: https://www.youtube.com/watch?v=LgOagKX_nA. Accessed July 26, 2021.
4. Elwyn G, Tilburt J, Montori VM. The ethical imperative for shared decision-making. *European Journal for Person Centered Healthcare*. 2013;1:129–213.
5. Stacey D, Legare F, Lewis K, et al. Decision aids for people facing health treatment or screening decisions. *Cochrane Database Syst Rev*. 2017;4:CD001431.
6. Hess EP. Clinical judgment is not passe when it comes to identifying patients with acute myocardial infarction. *Evid Based Med*. 2016;21:117.
7. Hess E. Authority Psychotherapy and the authority of the therapist in the religious Haredi community. *Am J Psychoanal*. 2018;78:137–158.
8. Hess EL, Myers EA, Swithers SE, et al. Associations between nonnutritive sweetener intake and metabolic syndrome in adults. *J Am Coll Nutr*. 2018;37:487–493.

4.2. Evaluation of Acute Chest Pain With Nonischemic Cardiac Pathologies

1. Labovitz AJ, Noble VE, Bierig M, et al. Focused cardiac ultrasound in the emergent setting: a consensus statement of the American Society of Echocardiography and American College of Emergency Physicians. *J Am Soc Echocardiogr*. 2010;23:1225–1230.
2. Roffi M, Patrono C, Collet JP, et al. 2015 ESC guidelines for the management of acute coronary syndromes in patients presenting without persistent ST-segment elevation: Task Force for the Management of Acute Coronary Syndromes in Patients Presenting Without Persistent ST-Segment Elevation of the European Society of Cardiology (ESC) *Eur Heart J*. 2016;37:267–315.
3. Amsterdam EA, Wenger NK, Brindis RG, et al. 2014 AHA/ACC guideline for the management of patients with non-ST-elevation acute coronary syndromes: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2014;130:e344–e426.
4. Douglas PS, Garcia MJ, Haines D, et al. ACCF/ASE/AHA/ASNC/HFSA/HRS/SCAI/SCCM/SCCT/SCMR 2011 appropriate use criteria for echocardiography: a report of the American College of Cardiology Foundation Appropriate Use Criteria Task Force, American Society of Echocardiography, American Heart Association, American Society of Nuclear Cardiology, Heart Failure Society of America, Heart Rhythm Society, Society for Cardiovascular Angiography and Interventions, Society of Critical Care Medicine, Society of Cardiovascular Computed Tomography, and Society for Cardiovascular Magnetic Resonance. *J Am Coll Cardiol*. 2011;57:1126–1166.
5. Sicari R, Nihoyannopoulos P, Evangelista A, et al. Stress echocardiography expert consensus statement—executive summary: European Association of Echocardiography (EAE) (a registered branch of the ESC) *Eur Heart J*. 2009;30:278–289.
6. Medical Advisory Secretariat. Stress echocardiography for the diagnosis of coronary artery disease: an evidence-based analysis. *Ont Health Technol Assess Ser*. 2010;10:1–61.

4.2.1. Acute Chest Pain With Suspected Acute Aortic Syndrome

1. Vilacosta I, San Roman JA. Acute aortic syndrome. *Heart*. 2001;85:365–368.
2. Benjamin EJ, Virani SS, Callaway CW, et al. Heart disease and stroke statistics—2018 update: a report from the American Heart Association. *Circulation*. 2018;137:e67–e492.
3. Spittell PC, Spittell JA Jr, Joyce JW, et al. Clinical features and differential diagnosis of aortic dissection: experience with 236 cases (1980 through 1990) *Mayo Clin Proc*. 1993;68:642–651.
4. Hiratzka LF, Bakris GL, Beckman JA, et al. 2010 ACCF/AHA/AATS/ACR/ASA/SCA/SCAI/SIR/STS/SVM guidelines for the diagnosis and management of patients with thoracic aortic disease: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines, American Association for Thoracic Surgery, American College of Radiology/American Stroke Association, Society of Cardiovascular Anesthesiologists, Society for Cardiovascular Angiography and Interventions, Society of Interventional Radiology, Society of Thoracic Surgeons, and Society for Vascular Medicine. *Circulation*. 2010;121:e266–e369.
5. Mussa FF, Horton JD, Moridzadeh R, et al. Acute aortic dissection and intramural hematoma: a systematic review. *JAMA*. 2016;316:754–763.
6. Vardhanabhuti V, Nicol E, Morgan-Hughes G, et al. Recommendations for accurate CT diagnosis of suspected acute aortic syndrome (AAS)—on behalf of the British Society of Cardiovascular Imaging (BSCI)/British Society of Cardiovascular CT (BSCCT) *Br J Radiol*. 2016;89:20150705.

4.2.2. Acute Chest Pain With Suspected PE

1. Pollack CV, Schreiber D, Goldhaber SZ, et al. Clinical characteristics, management, and outcomes of patients diagnosed with acute pulmonary embolism in the emergency department: initial report of EMPEROR (Multicenter Emergency Medicine Pulmonary Embolism in the Real World Registry) *J Am Coll Cardiol*. 2011;57:700–706.
2. Konstantinides SV, Torbicki A, Agnelli G, et al. 2014 ESC guidelines on the diagnosis and management of acute pulmonary embolism. *Eur Heart J*. 2014;35:3033–3069; 69a–69k.

3. Raja AS, Greenberg JO, Qaseem A, et al. Evaluation of patients with suspected acute pulmonary embolism: best practice advice from the Clinical Guidelines Committee of the American College of Physicians. *Ann Intern Med*. 2015;163:701–711.
4. Ceriani E, Combescurie C, Le Gal G, et al. Clinical prediction rules for pulmonary embolism: a systematic review and meta-analysis. *J Thromb Haemost*. 2010;8:957–970.
5. Alotaibi GS, Wu C, Senthilselvan A, et al. Secular trends in incidence and mortality of acute venous thromboembolism: the AB-VTE Population-Based Study. *Am J Med*. 2016;129:e19–e25.
6. Smith SB, Geske JB, Kathuria P, et al. Analysis of national trends in admissions for pulmonary embolism. *Chest*. 2016;150:35–45.
7. Cohen AT, Agnelli G, Anderson FA, et al. Venous thromboembolism (VTE) in Europe. The number of VTE events and associated morbidity and mortality. *Thromb Haemost*. 2007;98:756–764.
8. Kline JA, Jones AE, Shapiro NI, et al. Multicenter, randomized trial of quantitative pretest probability to reduce unnecessary medical radiation exposure in emergency department patients with chest pain and dyspnea. *Circ Cardiovasc Imaging*. 2014;7:66–73.

4.2.3. Acute Chest Pain With Suspected Myopericarditis

1. Bozkurt B, Colvin M, Cook J, et al. Current diagnostic and treatment strategies for specific dilated cardiomyopathies: a scientific statement from the American Heart Association. *Circulation*. 2016;134:e579–e646.
2. Ferreira VM, Schulz-Menger J, Holmvang G, et al. Cardiovascular magnetic resonance in nonischemic myocardial inflammation: expert recommendations. *J Am Coll Cardiol*. 2018;72:3158–3176.
3. Tornvall P, Gerbaud E, Behaghel A, et al. Myocarditis or "true" infarction by cardiac magnetic resonance in patients with a clinical diagnosis of myocardial infarction without obstructive coronary disease: a meta-analysis of individual patient data. *Atherosclerosis*. 2015;241:87–91.
4. Dastidar AG, Baritussio A, De Garate E, et al. Prognostic role of CMR and conventional risk factors in myocardial infarction with nonobstructed coronary arteries. *J Am Coll Cardiol Img*. 2019;12:1973–1982.
5. Lintings PF, Nivet H, Clement-Guinaudeau S, et al. High-resolution late gadolinium enhancement magnetic resonance for the diagnosis of myocardial infarction with nonobstructed coronary arteries. *J Am Coll Cardiol Img*. 2020;13:1135–1148.
6. Dastidar AG, Rodrigues JCL, Johnson TW, et al. Myocardial infarction with nonobstructed coronary arteries: impact of CMR early after presentation. *J Am Coll Cardiol Img*. 2017;10:1204–1206.
7. Klein AL, Abbasa S, Agler DA, et al. American Society of Echocardiography clinical recommendations for multimodality cardiovascular imaging of patients with pericardial disease. *J Am Soc Echocardiogr*. 2013;26:965–1012, e15.
8. Adler Y, Charron P, Imazio M, et al. 2015 ESC guidelines for the diagnosis and management of pericardial diseases: the Task Force for the Diagnosis and Management of Pericardial Diseases of the European Society of Cardiology (ESC) *Eur Heart J*. 2015;36:2921–2964.
9. Taylor AM, Dymarkowski S, Verbeken EK, et al. Detection of pericardial inflammation with late-enhancement cardiac magnetic resonance imaging: initial results. *Eur Radiol*. 2006;16:569–574.
10. Young PM, Glockner JF, Williamson EE, et al. MR imaging findings in 76 consecutive surgically proven cases of pericardial disease with CT and pathologic correlation. *Int J Cardiovasc Imaging*. 2012;28:1099–1109.
11. Aquaro GD, Perfetti M, Camastra G, et al. Cardiac MR with late gadolinium enhancement in acute myocarditis with preserved systolic function: ITAMY study. *J Am Coll Cardiol*. 2017;70:1977–1987.
12. Grani C, Buechel RR, Kaufmann PA, et al. Multimodality imaging in individuals with anomalous coronary arteries. *J Am Coll Cardiol Img*. 2017;10:471–481.
13. Hammer MM, Raptis CA, Javidan-Nejad C, et al. Accuracy of computed tomography findings in acute pericarditis. *Acta Radiol*. 2014;55:1197–1202.
14. Imazio M, Demicheli B, Cecchi E, et al. Cardiac troponin I in acute pericarditis. *J Am Coll Cardiol*. 2003;42:2144–2148.
15. Caforio AL, Pankuweit S, Arbustini E, et al. Current state of knowledge on aetiology, diagnosis, management, and therapy of myocarditis: a position statement of the European Society of Cardiology Working Group on Myocardial and Pericardial Diseases. *Eur Heart J*. 2013;34:2636–2648; 48a–48d.

16. Pathik B, Raman B, Mohd Amin NH, et al. Troponin-positive chest pain with unobstructed coronary arteries: incremental diagnostic value of cardiovascular magnetic resonance imaging. *Eur Heart J Cardiovasc Imaging*. 2016;17:1146–1152.
17. Cosyns B, Plein S, Nihoyanopoulos P, et al. European Association of Cardiovascular Imaging (EACVI) position paper: multimodality imaging in pericardial disease. *Eur Heart J Cardiovasc Imaging*. 2015;16:12–31.

4.2.4. Acute Chest Pain With Valvular Heart Disease

1. Morrison GW, Thomas RD, Grimmer SF, et al. Incidence of coronary artery disease in patients with valvular heart disease. *Br Heart J*. 1980;44:630–637.
2. Ahn JH, Kim SM, Park SJ, et al. Coronary microvascular dysfunction as a mechanism of angina in severe AS: prospective adenosine-stress CMR study. *J Am Coll Cardiol*. 2016;67:1412–1422.
3. Otto CM, Nishimura RA, Bonow RO, et al. 2020 ACC/AHA guideline for the management of patients with valvular heart disease: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2021;143:e72–e227.
4. Hiratzka LF, Bakris GL, Beckman JA, et al. 2010 ACCF/AHA/AATS/ACR/ASA/SCA/SCAI/SIR/STS/ SVM guidelines for the diagnosis and management of patients with thoracic aortic disease: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines, American Association for Thoracic Surgery, American College of Radiology, American Stroke Association, Society of Cardiovascular Anesthesiologists, Society for Cardiovascular Angiography and Interventions, Society of Interventional Radiology, Society of Thoracic Surgeons, and Society for Vascular Medicine. *Circulation*. 2010;121:e266–e369.
5. Lang RM, Badano LP, Tsang W, et al. EAE/ASE recommendations for image acquisition and display using three-dimensional echocardiography. *Eur Heart J Cardiovasc Imaging*. 2012;13:1–46.
6. Doherty JU, Kort S, Mehran R, et al. ACC/AATS/AHA/ASE/ASNC/HRS/SCAI/SCCT/SCMR/STS 2017 appropriate use criteria for multimodality imaging in valvular heart disease: a report of the American College of Cardiology Appropriate Use Criteria Task Force, American Association for Thoracic Surgery, American Heart Association, American Society of Echocardiography, American Society of Nuclear Cardiology, Heart Rhythm Society, Society for Cardiovascular Angiography and Interventions, Society of Cardiovascular Computed Tomography, Society for Cardiovascular Magnetic Resonance, and Society of Thoracic Surgeons. *J Am Coll Cardiol*. 2017;70:1647–1672.

4.3. Evaluation of Acute Chest Pain With Suspected Noncardiac Causes

4.3.1. Evaluation of Acute Chest Pain With Suspected Gastrointestinal Syndromes

1. Klinkman MS, Stevens D, Gorenflo DW. Episodes of care for chest pain: a preliminary report from MIRNET. Michigan Research Network. *J Fam Pract*. 1994;38:345–352.
2. Sengupta JN. An overview of esophageal sensory receptors. *Am J Med*. 2000;108(suppl 4a):87S–89S.
3. Dent J, El-Serag HB, Wallander MA, et al. Epidemiology of gastro-oesophageal reflux disease: a systematic review. *Gut*. 2005;54:710–717.
4. Richter JE. Typical and atypical presentations of gastroesophageal reflux disease. The role of esophageal testing in diagnosis and management. *Gastroenterol Clin North Am*. 1996;25:75–102.
5. DeVault KR, Castell DO. Guidelines for the diagnosis and treatment of gastroesophageal reflux disease. Practice Parameters Committee of the American College of Gastroenterology. *Arch Intern Med*. 1995;155:2165–2173.
6. Hirano I, Richter JE. Practice parameters committee of the American College of Gastroenterology. ACG practice guidelines: esophageal reflux testing. *Am J Gastroenterol*. 2007;102:668–685.

4.3.2. Evaluation of Acute Chest Pain With Suspected Anxiety and Other Psychosomatic Considerations

1. Carter C, Maddock R, Amsterdam E, et al. Panic disorder and chest pain in the coronary care unit. *Psychosomatics*. 1992;33:302–309.

2. Kline JA, Shapiro NI, Jones AE, et al. Outcomes and radiation exposure of emergency department patients with chest pain and shortness of breath and ultralow pretest probability: a multicenter study. *Ann Emerg Med*. 2014;63:281–288.
3. Webster R, Norman P, Goodacre S, et al. The prevalence and correlates of psychological outcomes in patients with acute non-cardiac chest pain: a systematic review. *Emerg Med J*. 2012;29:267–273.
4. Eslick GD, Talley NJ. Natural history and predictors of outcome for non-cardiac chest pain: a prospective 4-year cohort study. *Neurogastroenterol Motil*. 2008;20:989–997.
5. Foldes-Busque G, Marchand A, Chauny JM, et al. Unexplained chest pain in the ED: could it be panic? *Am J Emerg Med*. 2011;29:743–751.
6. Musey PI Jr, Kline JA. Emergency department cardiopulmonary evaluation of low-risk chest pain patients with self-reported stress and anxiety. *J Emerg Med*. 2017;52:273–279.
7. Al-Ani M, Winchester DE. Prevalence and overlap of noncardiac conditions in the evaluation of low-risk acute chest pain patients. *Crit Pathw Cardiol*. 2015;14:97–102.
8. Czarnecki A, Wang JT, Tu JV, et al. The role of primary care physician and cardiologist follow-up for low-risk patients with chest pain after emergency department assessment. *Am Heart J*. 2014;168:289–295.
9. White KS, Craft JM, Gervino EV. Anxiety and hypervigilance to cardiopulmonary sensations in non-cardiac chest pain patients with and without psychiatric disorders. *Behav Res Ther*. 2010;48:394–401.
10. Burgstaller JM, Jenni BF, Steurer J, et al. Treatment efficacy for non-cardiovascular chest pain: a systematic review and meta-analysis. *PLoS One*. 2014;9:e104722.
11. Kisely SR, Campbell LA, Yelland MJ, et al. Psychological interventions for symptomatic management of non-specific chest pain in patients with normal coronary anatomy. *Cochrane Database Syst Rev*. 2015;CD004101.
12. Mitchell AM, Garvey JL, Chandra A, et al. Prospective multicenter study of quantitative pretest probability assessment to exclude acute coronary syndrome for patients evaluated in emergency department chest pain units. *Ann Emerg Med*. 2006;47:447.
13. Hoorweg BB, Willemsen RT, Cleef LE, et al. Frequency of chest pain in primary care, diagnostic tests performed and final diagnoses. *Heart*. 2017;103:1727–1732.
14. Glombiewski JA, Rief W, Bosner S, et al. The course of nonspecific chest pain in primary care: symptom persistence and health care usage. *Arch Intern Med*. 2010;170:251–255.
15. Engel GL. Sudden and rapid death during psychological stress. Folklore or folk wisdom? *Ann Intern Med*. 1971;74:771–782.
16. Samuels MA. The brain-heart connection. *Circulation*. 2007;116:77–84.
17. Rudehill A, Olsson GL, Sundqvist K, et al. ECG abnormalities in patients with subarachnoid haemorrhage and intracranial tumours. *J Neurol Neurosurg Psychiatry*. 1987;50:1375–1381.
18. Wittstein IS, Thieman DR, Lima JA, et al. Neurohumoral features of myocardial stunning due to sudden emotional stress. *N Engl J Med*. 2005;352:539–548.
19. Medina de Chazal H, Del Buono MG, Keyser-Marcus L, et al. Stress cardiomyopathy diagnosis and treatment: JACC state-of-the-art review. *J Am Coll Cardiol*. 2018;72:1955–1971.
20. Bass C, Wade C, Hand D, et al. Patients with angina with normal and near normal coronary arteries: clinical and psychosocial state 12 months after angiography. *Br Med J (Clin Res Ed)*. 1983;287:1505–1508.
21. Mukerji V, Beitman BD, Alpert MA. Chest pain and angiographically normal coronary arteries. Implications for treatment. *Tex Heart Inst J*. 1993;20:170–179.
22. Lichtlen PR, Bargheer K, Wenzlaff P. Long-term prognosis of patients with anginalike chest pain and normal coronary angiographic findings. *J Am Coll Cardiol*. 1995;25:1013–1018.
23. Esler JL, Barlow DH, Woolard RH, et al. A brief cognitive-behavioral intervention for patients with noncardiac chest pain. *Behavior Therapy*. 2003;34:129–148.
24. Aikens JE, Zvolensky MJ, Eifert GH. Differential fear of cardiopulmonary sensations in emergency room noncardiac chest pain patients. *J Behav Med*. 2001;24:155–167.
25. Eifert GH, Zvolensky MJ, Lejuez CW. Heart-focused anxiety and chest pain: a conceptual and clinical review. *Clinical Psychology Science and Practice*. 2006;7:403–417.
26. Esler JL, Bock BC. Psychological treatments for noncardiac chest pain: recommendations for a new approach. *J Psychosom Res*. 2004;56:263–269.

27. Altintas E, Yigit F, Taskintuna N. The impact of psychiatric disorders with cardiac syndrome X on quality of life: 3 months prospective study. *Int J Clin Exp Med*. 2014;7:3520–3527.
28. Schwarz J, Prashad A, Winchester DE. Prevalence and implications of severe anxiety in a prospective cohort of acute chest pain patients. *Crit Pathw Cardiol*. 2015;14:44–47.
29. Folds-Busque G, Denis I, Poitras J, et al. A closer look at the relationships between panic attacks, emergency department visits and non-cardiac chest pain. *J Health Psychol*. 2017; 1359105316683785.
30. Campbell KA, Madva EN, Villegas AC, et al. Non-cardiac chest pain: a review for the consultation-liaison psychiatrist. *Psychosomatics*. 2017;58:252–265.

4.3.3. Evaluation of Acute Chest Pain in Patients With Sickle Cell Disease

1. Platt OS, Brambilla DJ, Rosse WF, et al. Mortality in sickle cell disease. Life expectancy and risk factors for early death. *N Engl J Med*. 1994;330:1639–1644.
2. Thomas AN, Pattison C, Serjeant GR. Causes of death in sickle-cell disease in Jamaica. *Br Med J (Clin Res Ed)*. 1982;285:633–635.
3. Vichinsky EP, Styles LA, Colangelo LH, et al. Acute chest syndrome in sickle cell disease: clinical presentation and course. Cooperative Study of Sickle Cell Disease. *Blood*. 1997;89:1787–1792.
4. Martin CR, Johnson CS, Cobb C, et al. Myocardial infarction in sickle cell disease. *J Natl Med Assoc*. 1996;88:428–432.
5. Ogunbayo GO, Misumida N, Olorunfemi O, et al. Comparison of outcomes in patients having acute myocardial infarction with versus without sickle-cell anemia. *Am J Cardiol*. 2017;120:1768–1771.

5. EVALUATION OF PATIENTS WITH STABLE CHEST PAIN

5.1. Patients With No Known CAD Presenting With Stable Chest Pain

1. Juarez-Orozco LE, Saraste A, Capodanno D, et al. Impact of a decreasing pre-test probability on the performance of diagnostic tests for coronary artery disease. *Eur Heart J Cardiovasc Imaging*. 2019;20:1198–1207.
2. Winther S, Schmidt SE, Mayrhofer T, et al. Incorporating coronary calcification into pre-test assessment of the likelihood of coronary artery disease. *J Am Coll Cardiol*. 2020;76:2421–2432.

5.1.1. Pretest Risk Probability to Guide Need for Stress and Anatomic Tests

1. Juarez-Orozco LE, Saraste A, Capodanno D, et al. Impact of a decreasing pre-test probability on the performance of diagnostic tests for coronary artery disease. *Eur Heart J Cardiovasc Imaging*. 2019;20:1198–1207.
2. Winther S, Schmidt SE, Mayrhofer T, et al. Incorporating coronary calcification into pre-test assessment of the likelihood of coronary artery disease. *J Am Coll Cardiol*. 2020;76:2421–2432.
3. Genders TS, Steyerberg EW, Alkadhi H, et al. A clinical prediction rule for the diagnosis of coronary artery disease: validation, updating, and extension. *Eur Heart J*. 2011;32:1316–1330.
4. Fordyce CB, Douglas PS, Roberts RS, et al. Identification of patients with stable chest pain deriving minimal value from noninvasive testing: the PROMISE minimal-risk tool, a secondary analysis of a randomized clinical trial. *JAMA Cardiol*. 2017;2:400–408.
5. Genders TS, Steyerberg EW, Hunink MG, et al. Prediction model to estimate presence of coronary artery disease: retrospective pooled analysis of existing cohorts. *BMJ*. 2012;344:e3485.

5.1.2. Low-Risk Patients With Stable Chest Pain and No Known CAD

1. Genders TS, Steyerberg EW, Alkadhi H, et al. A clinical prediction rule for the diagnosis of coronary artery disease: validation, updating, and extension. *Eur Heart J*. 2011;32:1316–1330.
2. Fordyce CB, Douglas PS, Roberts RS, et al. Identification of patients with stable chest pain deriving minimal value from noninvasive testing: the

PROMISE minimal-risk tool, a secondary analysis of a randomized clinical trial. *JAMA Cardiol*. 2017;2:400–408.

3. Genders TS, Steyerberg EW, Hunink MG, et al. Prediction model to estimate presence of coronary artery disease: retrospective pooled analysis of existing cohorts. *BMJ*. 2012;344:e3485.
4. Juarez-Orozco LE, Saraste A, Capodanno D, et al. Impact of a decreasing pre-test probability on the performance of diagnostic tests for coronary artery disease. *Eur Heart J Cardiovasc Imaging*. 2019;20:1198–1207.
5. Winther S, Schmidt SE, Mayrhofer T, et al. Incorporating coronary calcification into pre-test assessment of the likelihood of coronary artery disease. *J Am Coll Cardiol*. 2020;76:2421–2432.
6. Lubbers M, Coenen A, Kofflard M, et al. Comprehensive cardiac CT with myocardial perfusion imaging versus functional testing in suspected coronary artery disease: the multicenter, randomized CRESCENT-II trial. *J Am Coll Cardiol Img*. 2018;11:1625–1636.
7. Lubbers M, Dedic A, Coenen A, et al. Calcium imaging and selective computed tomography angiography in comparison to functional testing for suspected coronary artery disease: the multicenter, randomized CRESCENT trial. *Eur Heart J*. 2016;37:1232–1243.
8. Chang SM, Nabi F, Xu J, et al. The coronary artery calcium score and stress myocardial perfusion imaging provide independent and complementary prediction of cardiac risk. *J Am Coll Cardiol*. 2009;54:1872–1882.
9. Budoff MJ, Mayrhofer T, Ferencik M, et al. Prognostic value of coronary artery calcium in the PROMISE study (prospective multicenter imaging study for evaluation of chest pain). *Circulation*. 2017;136:1993–2005.
10. Shaw LJ, Mieres JH, Hendel RH, et al. Comparative effectiveness of exercise electrocardiography with or without myocardial perfusion single photon emission computed tomography in women with suspected coronary artery disease: results from the What Is the Optimal Method for Ischemia Evaluation in Women (WOMEN) trial. *Circulation*. 2011;124:1239–1249.
11. Douglas PS, Hoffmann U. Anatomical versus functional testing for coronary artery disease. *N Engl J Med*. 2015;373:91.
12. Hochman JS, Reynolds HR, Bangalore S, et al. Baseline characteristics and risk profiles of participants in the ISCHEMIA randomized clinical trial. *JAMA Cardiol*. 2019;4:273–286.
13. Spertus JA, Jones PG, Maron DJ, et al. Health status after invasive or conservative care in coronary and advanced kidney disease. *N Engl J Med*. 2020;382:1619–1628.
14. Rozanski A, Gransar H, Hayes SW, et al. Temporal trends in the frequency of inducible myocardial ischemia during cardiac stress testing: 1991 to 2009. *J Am Coll Cardiol*. 2013;61:1054–1065.
15. Cheng VY, Berman DS, Rozanski A, et al. Performance of the traditional age, sex, and angina typicality-based approach for estimating pretest probability of angiographically significant coronary artery disease in patients undergoing coronary computed tomographic angiography: results from the multinational coronary CT angiography evaluation for clinical outcomes: an international multicenter registry (CONFIRM). *Circulation*. 2011;124:2423–2432; 1–8.
16. Baskaran L, Danad I, Gransar H, et al. A comparison of the updated Diamond-Forrester, CAD Consortium, and CONFIRM history-based risk scores for predicting obstructive coronary artery disease in patients with stable chest pain: the SCOT-HEART Coronary CTA cohort. *J Am Coll Cardiol Img*. 2019;12:1392–1400.
17. Ferreira AM, Marques H, Tralhao A, et al. Pre-test probability of obstructive coronary stenosis in patients undergoing coronary CT angiography: comparative performance of the modified diamond-Forrester algorithm versus methods incorporating cardiovascular risk factors. *Int J Cardiol*. 2016;222:346–351.
18. Foldyna B, Udelson JE, Karady J, et al. Pretest probability for patients with suspected obstructive coronary artery disease: re-evaluating Diamond-Forrester for the contemporary era and clinical implications: insights from the PROMISE trial. *Eur Heart J Cardiovasc Imaging*. 2019;20:574–581.
19. Bittencourt MS, Hultén E, Polonsky TS, et al. European Society of Cardiology-recommended coronary artery disease consortium pretest probability scores more accurately predict obstructive coronary disease and cardiovascular events than the Diamond and Forrester score: the Partners registry. *Circulation*. 2016;134:201–211.
20. Patel MR, Peterson ED, Dai D, et al. Low diagnostic yield of elective coronary angiography. *N Engl J Med*. 2010;362:886–895.
21. Patel MR, Dai D, Hernandez AF, et al. Prevalence and predictors of non-obstructive coronary artery disease identified with coronary angiography in contemporary clinical practice. *Am Heart J*. 2014;167:846–852 e2.
22. Knuuti J, Wijns W, Saraste A, et al. 2019 ESC guidelines for the diagnosis and management of chronic coronary syndromes. *Eur Heart J*. 2020;41:407–477.